Bayesian optimization with GPU acceleration for ocean models

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Key Points:

- Bayesian optimization is used to reduce mixed layer depth biases in a primitive equation 1° ocean model.
- The JAX library allows us to use GPUs and reduce energy consumption by more than 90%.
- The tool can be easily used for optimization of other models.

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Abstract

Ocean general circulation models (OGCMs) contain numerous parameterizations of subgrid scale processes. The parameter tuning procedure is rarely reported and often done by hand. We present an automated alternative: VerOpt, a Python package for the ocean model Veros, adapts Bayesian optimization to climate model tuning. We use VerOpt to identify a set of model parameter values that minimize mixed layer depth (MLD) bias in Veros. We present the results of three optimization procedures: TWIN, OBS_TKE and OBS_9D. The goal is to minimize MLD error relative to a target. In TWIN, the target is MLD simulated using Veros with a known vertical mixing parameterization. VerOpt identifies a set of parameter values that simulate the target MLD with 99% accuracy. In OBS_TKE and OBS_9D, the target is MLD climatology. In the OBS_TKE optimization, only the vertical turbulence closure parameters are calibrated, which is insufficient to reduce MLD biases in Veros. In the OBS_9D experiment, the mesoscale eddy closure and the scaling of the wind stress and salinity forcing masks are tuned in addition to the vertical mixing scheme parameters. MLD biases in Veros are reduced by up to 23% with the largest impact in the Southern Ocean. In the ensemble of the simulations for which MLD biases are minimized, the mesoscale turbulence diffusion coefficients K_{gm} and K_{iso} are halved compared to the control run, and the zonal wind stress is 10% weaker.

Plain Language Summary

Numerous schemes in ocean models approximate the impact of unresolved processes on the mean state of the system. It is not always possible to determine the values of the model parameters based on empirical arguments. Often, the schemes are instead calibrated by hand to reproduce the most realistic ocean state. We present an alternative which automates the tuning process. We use Bayesian optimization (BO) to determine which parameter values minimize the mixed layer depth (MLD) biases in the ocean model Veros. The Versatile Optimizer (VerOpt) adapts BO to climate modeling problems. We show the results of three optimization sequences: TWIN, OBS_TKE and OBS_9D. In the TWIN experiment, the optimizer is tasked with finding a set of parameters which reproduce MLD simulated with Veros. VerOpt identifies a range of parameter values which reproduce the target MLD up to 1% accuracy. In the OBS_TKE experiment, the target is the observed MLD. We find that tuning the vertical mixing scheme parameters in isolation does not yield a reduction in MLD biases. We achieve this instead by including the horizontal mixing scheme and atmospheric forcing parameters in the OBS_9D optimization, where the model biases are reduced by up to 23%.

1 Introduction

Accurate mixed layer depth (MLD) representation is necessary for realistic energy exchange between the ocean and atmospheric components in coupled models, as biases in MLD lead to sea surface temperature (SST) anomalies. In the tropical ocean, SST anomalies can lead to restructuring of the global climate (Jochum & Potemra, 2008). Despite the community effort spanning multiple decades, MLD biases still persist in ocean general circulation models (OGCMs; Huang et al., 2014; Treguier et al., 2023). Closure schemes that represent unresolved sub-grid scale processes are a primary source of uncertainty in OGCMs (Fox-Kemper et al., 2019).

Only some of the parameter values in these schemes have a solid physical basis. For example, in the Turbulent Kinetic Energy (TKE) closure in Gaspar et al. (1990), the coefficients c_k and c_ϵ are picked based on the empirical value of the mixing coefficient γ_{R_f}, which is highly uncertain (Gregg et al., 2018). The OGCM parameters that need to be selected by the modeler are often tuned by hand, and the process is rarely reported (Mauritsen et al., 2012; Hourdin et al., 2017). Here, *tuning* refers to changing the model parameters such that the model output resembles observations as closely as possible. This process carries with itself a number of risks, such as over-fitting and bias compensation, highlighting the need for more systematic and automated methods for calibrating OGCMs.

We demonstrate such a method here by tuning the ocean model Veros (Versatile Ocean Simulator, Häfner et al., 2018, 2021) to better represent the annual mean mixed layer depth (MLD). We use the Python package VerOpt (Versatile Optimizer, Stoustrup, 2021), which is designed to adapt Bayesian optimization (BO) to climate modeling problems. In recent decades, BO has become a popular method to handle expensive black box optimization and has been successfully applied in disciplines such as robotics, environmental sensing, drug design, as well as in machine learning for hyper-parameter optimization (Shahriari et al., 2015; Wang et al., 2023). The method's popularity has catalyzed its development, as well as software availability. In Python, BO models can be constructed using BoTorch (Balandat et al., 2020). VerOpt is built in the PyTorch ecosystem (Ansel et al., 2024) and implements BoTorch optimization routines and functions.

We present the results of three optimization experiments: TWIN, OBS_TKE and OBS_9D. Each experiment uses VerOpt to select parameters that yield the closest match to the target MLD distribution. The optimization process and the fundamentals of BO are introduced in section 2. The setup of the three optimization procedures is described in section 3, including the model description, the choice of MLD climatology (section 3.1), the optimizer specifications (sections 3.2 and 3.3), and the computational resources (section 3.4). The results of the three optimization experiments are shown in section 4. The discussion in section 5 explores possible improvements and adaptations of the optimization procedure, followed by conclusions in section 6.

2 Bayesian optimization with VerOpt

Bayesian optimization is a black box optimization method, i.e. it locates the global minimum of an unknown scalar objective function $f(\mathbf{x})\colon \mathcal{X}^d \to \mathbb{R}$. Two aspects of the method make it especially well suited for climate science problems. Firstly, the optimization does not rely on the gradient of the objective function. A full simulation spanning millions of iterations is difficult to differentiate, and for most existing earth system models it is completely out of reach. Furthermore, the gradients may contain singularities or bifurcations, which break the optimization sequence. Secondly, BO requires relatively few objective function evaluations. Optimization algorithms that do not rely on gradient information often depend on tens of thousands to millions of input points. Running such vast numbers of OGCM simulations is unfeasible. Efficient sampling is therefore a top priority in OGCM calibration.

The data efficiency of BO stems from utilizing Gaussian process (GP) regression to construct a model of $f(\mathbf{x})$ over $\mathcal{X}$. The so-called *surrogate* of the objective function is then used to compute the *acquisition* function, which highlights the regions in the parameter space that are most likely to contain the global objective function minimum:

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathcal{X}^d} f(\mathbf{x}).^1 \tag{1}$$

The objective function is sampled iteratively following the three key steps of the optimization procedure:

1. the construction of the surrogate model, which emulates the distribution of possible objective functions given our existing knowledge,

[1] VerOpt follows the Bayesian optimization literature custom where the objective function is *maximized* rather than minimized. However, since the aim of the optimization in this paper is to minimize model biases, we present it as such for the sake of readability and consistency.

2. the calculation of the acquisition function from the mean and uncertainty of the surrogate model, and
3. the minimization of the acquisition function to determine the new objective function coordinates to evaluate.

The steps involved in optimizing an unknown function $f\colon \mathcal{X} \to \mathbb{R}$ with VerOpt are summarized in Figure 1. The following description of the method is highly condensed; more in-depth explanations can be found in Stoustrup (2021) and Williams and Rasmussen (2006).

Let us consider a set of objective function evaluations: $\{(\mathbf{x}_i, f_i)|i = 1, \ldots, n\}$. Here, $\mathbf{x}_i$ is an input point in the parameter space $\mathcal{X}^d$ with d parameters. The $d \times n$ matrix X contains all n input points. The corresponding objective function values $f_i(\mathbf{x}_i)$ are contained in the vector $\mathbf{f}(X) \in \mathbb{R}^n$. GP regression allows us to construct a predictive distribution of the surrogate model $\mathbf{f}_*$ at a set of test points X_*:

$$\begin{aligned} p(\mathbf{f}_*(X_*)|X_*, \mathbf{f}(X)) = \mathcal{N}(&K(X_*, X)K(X, X)^{-1}\mathbf{f}, \\ &K(X_*, X_*) - K(X_*, X)K(X, X)^{-1}K(X, X_*)). \end{aligned} \tag{2}$$

$\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ symbolizes the normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix Σ. We assume noise-free input $y_i = f_i$. The Gram matrices $K(X, X')$ contain covariances between any two set of points X and X', such that each element of the matrix can be described as $k_{ij} = k(\mathbf{x}_i, \mathbf{x}'_j)$. For example, an element of the covariance matrix between the test and input points, $K(X_*, X)$, is equal to $k(\mathbf{x}_{*i}, \mathbf{x}_j)$.

The *kernel* $k(\mathbf{x}, \mathbf{x}')$ determines the shape and behavior of the surrogate model; it contains the *prior* information about the objective function. The Matérn kernel is the default VerOpt option:

$$k(\mathbf{x}, \mathbf{x}') = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{||\mathbf{x} - \mathbf{x}'||}{l} \right)^{\nu} K_\nu \left(\sqrt{2\nu} \frac{||\mathbf{x} - \mathbf{x}'||}{l} \right). \tag{3}$$

$\Gamma(\cdot)$ is the gamma function, K_ν is the modified Bessel function of the order ν, and l is the characteristic length scale. The Matérn kernel has high utility across a wide range of black box optimization problems because:

1. It is $\lceil \nu \rceil - 1$ differentiable. Climate model output, just as most physical systems, is not expected to be infinitely-differentiable and smooth, so the hyper-parameter ν provides control over an important aspect of the objective function surrogate.
2. Aside from ν, the kernel contains only one other hyper-parameter - the characteristic length scale l, which can be determined by maximizing the marginal likelihood in Eq. (4).

In VerOpt, the default choice for ν is 2.5, making the kernel twice-differentiable. The optimal length scale is determined by maximizing the marginal log likelihood (MLL) of the surrogate model:

$$\log(p(\mathbf{f}(X)|\boldsymbol{\theta})) = -\frac{1}{2}\mathbf{f}^{\mathrm{T}} K(X, X)\mathbf{f} - \frac{1}{2} \log |K(X, X)| - \frac{n}{2} \log(2\pi), \tag{4}$$

where $\boldsymbol{\theta}$ is the hyper-parameter vector. In our case, $\boldsymbol{\theta} = \mathbf{l} = [l_1, \ldots, l_d]$. The anisotropic Matérn kernel allows the length scale parameter l to be different for each dimension in $\mathcal{X}$. The radial distance $r = \frac{||\mathbf{x}-\mathbf{x}'||}{l}$ in Eq. (3) becomes $r = \left|\left| \frac{\mathbf{x}-\mathbf{x}'}{\mathbf{l}} \right|\right|$.

The three terms in the MLL equation represent the model fit, the model complexity, and the normalization factor, respectively. The term $-\frac{1}{2}\mathbf{f}^{\mathrm{T}} K(X, X)\mathbf{f}$ measures how well the surrogate model covariance fits with the variance of the objective function values. At large length scales, the GP model becomes approximately constant, while low

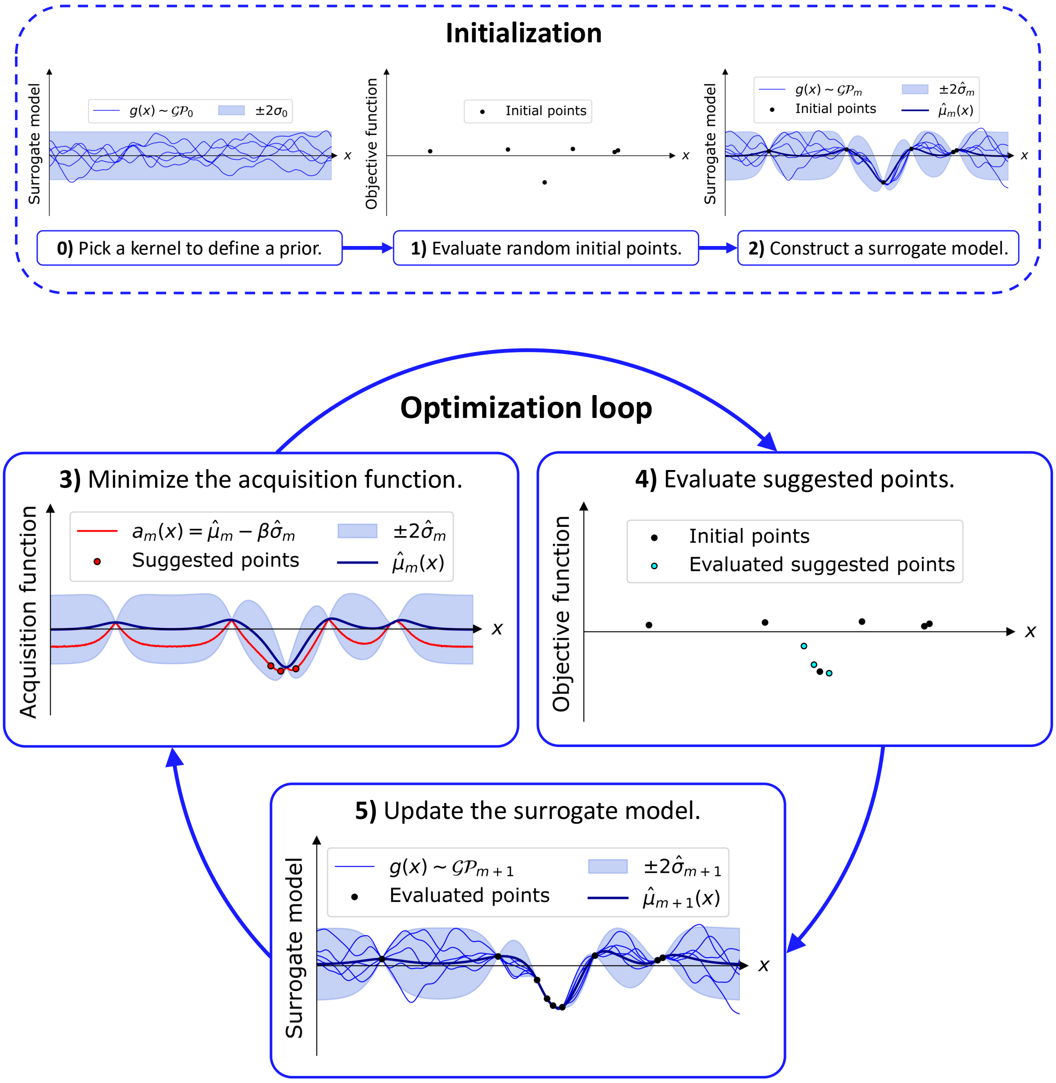

Figure 1. Bayesian optimization with VerOpt demonstrated using a one-dimensional example. The prior is a Gaussian process $\mathcal{GP}_0(0, k(x, x'))$, where $k(x, x')$ is the Matérn kernel. Five sample functions $g_{j\in\{1,\ldots,5\}}(x) \sim \mathcal{GP}_0$ illustrate our assumption about the shape of the objective function. The shaded region indicates the 95% confidence bound. In panel 1), the optimization is initialized by evaluating $n_{init} = 6$ random objective function coordinates. In the first optimization round $m = 1$, the surrogate model is constructed based on the initial points (panel 2). For $m \in \{1, \ldots, n_{rounds}\}$, the acquisition function $a_m(x)$ with $\beta = 1$ is calculated from the surrogate model mean $\hat{\mu}_m$ and uncertainty $\hat{\sigma}_m$ (panel 3). The acquisition function minima are suggested to evaluate. Panel 4 shows the updated set of objective function values: the basis for the updated surrogate model $\mathcal{GP}_{m+1}$ in panel 5. Each optimization round consists of the steps shown in panels 3-5.

length scales produce highly nonlinear models. The punishment term for model complexity $\frac{1}{2}\log|K(X,X)|$ is therefore high at low length scales where the surrogate is flexible and can fit the evaluated points well. The bad fit punishment $\frac{1}{2}\mathbf{f}^{\mathrm{T}}K(X,X)\mathbf{f}$ grows exponentially as l increases. This makes the MLL convex and the maximum easy to find.

After the optimal hyper-parameter vector $\hat{\boldsymbol{\theta}}$ is determined, the predictive distribution of the surrogate model $p(\mathbf{f}_*)$ is used to compute the acquisition function. The default VerOpt acquisition function is the modified Lower Confidence Bound (LCB, Srinivas et al., 2009):

$$a(\mathbf{x}) = \hat{\mu}(\mathbf{x}) - \hat{\sigma}(\mathbf{x})\beta - r\hat{\sigma}(\mathbf{x})\beta\gamma, \tag{5}$$

where β and γ are positive acquisition function hyper-parameters, $\hat{\mu}$ and $\hat{\sigma}$ are the mean and standard deviation of the optimal surrogate model, and r is a random number drawn from $\mathcal{N}(0,1)$. The hyper-parameter β value determines whether the optimizer should prioritize *exploitation* or *exploration*. Higher β values lead to more exploration, meaning that the optimizer searches the uncertain regions of the parameter space. Conversely, low β values lead to more exploitation of the surrogate model, making the optimizer search the regions where $\hat{\mu}$ is smallest. In VerOpt, $\beta = 3$ by default. The standard form of LCB does not include the γ term. The stochastic noise is added to improve optimizer performance when the surrogate model is nearly constant along a parameter. The value of $\gamma = 0.01$ is sufficiently small to only affect the optimization in that case.

The acquisition function is optimized using the L-BFGS-B algorithm with multiple restarts (Limited-memory Broyden-Fletcher-Goldfarb-Shanno with Box constraints, Byrd et al., 1995), as implemented in the scikit-learn package (Pedregosa et al., 2011). The new set of the optimal coordinates is evaluated by the objective function, i.e. by running a batch of simulations with parameterizations $\{\mathbf{x}_1, \ldots, \mathbf{x}_{n\_evals}\}$. If the batch consists of more than one simulation, $n_{evals} > 1$, an exponentially decaying punishment $p(\mathbf{x})$ is added to the acquisition function around the already selected objective function coordinates to avoid repeatedly evaluating the same or very similar parameter values:

$$p(\mathbf{x}) = -\sum_j a(\mathbf{x})\omega \exp\left(-\frac{|\mathbf{x}-\mathbf{x}_j|}{\alpha^2}\right), \tag{6}$$

where $\mathbf{x}_j$ is the j^{th} coordinate selected in the i^{th} optimization round, $j \in \{1, \ldots, n_{evals}\}$, and ω and α are hyper-parameters equal to unity by default.

In total, the optimization produces n objective function evaluations, where $n = n_{init} + n_{Bayes}$. The n_{init} initial points are randomly picked from a uniform distribution over the finite bounds of the parameter space $\mathcal{X}$. The n_{Bayes} points are the minima of the acquisition function. At each round of optimization, n_{evals} points are evaluated. The total number of optimization rounds is therefore $n_{rounds} = n_{Bayes}/n_{evals}$. The evaluated set of parameter values for which the objective function is minimized is denoted by $\boldsymbol{\chi}^*$.

In the optimization sequence in Figure 1, the task is to determine the parameter x for which $f(x)$ is minimized. Panel 0 shows five random vector draws from the GP prior $\mathcal{GP}_0(0, k_0(x,x'))$, where $k_0(x,x')$ is the Matérn kernel with $\nu = 2.5$ and $l_0 = 0.25$ (Eq. (3)). The shaded region indicates $\mu_0(x) \pm 2\sigma_0(x)$. The optimization is initialized by evaluating n_{init} randomly selected initial points (panel 1). The GP prior distribution is conditioned on the evaluated points (Eq. (2)) to construct the first surrogate model at optimization round $m = 1$. The optimal length scale $\hat{l}_1$ is found by locating the MLL maximum from Eq. (3). The draws from $\mathcal{GP}_1$ show the possible shape of the objective function given the information gained from evaluating the initial points. In panel 3, the acquisition function $a(x)$ (Eq. (5)) computed using the mean and variance from the surrogate model is plotted. The result of the acquisition function optimization is the new set of suggested points: the optimizer's best guesses for $\mathbf{x}^*$. In panel 4, the new evaluated points are plotted alongside the initial points. The full set is used to update the surrogate model to $\mathcal{GP}_{m+1}$ shown in panel 5. Each optimization round m produces:

- the surrogate model $\mathcal{GP}_m(\hat{\mu}_m(x), k_m(x, x'))$ with the kernel defined using the optimal length scale $\hat{l}_m$,
- the surrogate model mean $\hat{\mu}_m(x)$ and standard deviation $\hat{\sigma}_m(x)$,
- the acquisition function $a_m(x)$ and
- the set of points to evaluate $\{x_{m,1}, \ldots, x_{m,n\_evals}\}$.

Panels 3-5 summarise the steps of the optimization procedure repeated for $m \in \{1, \ldots, n_{rounds}\}$.

3 Description of the optimization routines

Two Veros setups are used in the optimization routines: the coarse setup with 4°×4° horizontal resolution (Veros 4°), and the standard setup with 1°×1° horizontal resolution (Veros 1°). The model employs a regular, three-dimensional staggered Arakawa-C grid. Both setups have 60 vertical layers with spacing monotonously increasing towards the bottom. In Veros 1°, the surface vertical resolution is 2 m, while in Veros 4° it is 4 m. The models are forced with monthly ERA-Interim (ECMWF Reanalysis v4, Dee et al., 2011) climatology. Further description of Veros components can be found in Häfner et al. (2021).

The duration of the simulations in TWIN and OBS_TKE is 30 model years, which is long enough for the tropical and mid-latitude mixed layer to spin up (Liu et al., 1994). In OBS_9D, because we extend the meridional bound of the experiment to 60°S, the duration of the model runs is set to 60 years. The integration begins from initial conditions based on climatological temperature and salinity profiles from World Ocean Atlas 2005 (WOA05, Locarnini et al., 2006; Antonov et al., 2006). Model evaluation is based on the mean of the final two years for TWIN and OBS_TKE, and the final decade for OBS_9D.

The vertical mixing scheme implemented in Veros is the Turbulent Kinetic Energy (TKE, Gaspar et al., 1990) closure with four free parameters: c_k, c_ϵ, α_{tke} and min_{K_m}. The prognostic turbulent kinetic energy ($\bar{e}$) equation is solved in the model to compute the vertical eddy viscosity and diffusivity, K_m and K_h, respectively. The parameter c_k scales the proportion of $\bar{e}$ used for mixing, and the parameter c_ϵ scales the dissipation of $\bar{e}$. The ratio of $c_k c_\epsilon^{-1}$ is proportional to the critical Richardson number (Ri_c). Vertical turbulent transport of $\bar{e}$ is parameterized as diffusive flux and scaled with α_{tke}. In the scheme, a minimum of $\bar{e}$ or of min_{K_m} can be set to impose background mixing below the boundary layer. The min_{K_m} method is the default pick in Veros, with the turbulent Prandtl number P_{rt} set to 10 in the interior ocean, such that $min_{K_m} = 10\, min_{K_h}$.

Mesoscale turbulence is parameterized through isopycnal diffusion of tracers (Redi, 1982) and horizontal diffusion of available potential energy (Gent et al., 1995). Each is proportional to a diffusion coefficient (K_{iso} and K_{gm}, respectively), which in the present set-up is equal for both and based on a closure for mesoscale turbulent energy (Eden & Greatbatch, 2008). Two key parameters impacting the K_{gm} value are eke_c_k and eke_c_ϵ. Similarly to the TKE closure, eke_c_k scales the K_{gm} coefficient and eke_c_ϵ scales the eddy kinetic energy dissipation. The TKE and EKE parameterizations are described in more detail in Appendix A.

The ocean surface in Veros is forced with radiative fluxes, wind stress and a surface salinity mask. It is thus not possible to directly optimize the bulk formula coefficients; instead, we implement three parameters that scale the forcing masks. The zonal and meridional wind stress fields are scaled by c_{τ_x} and c_{τ_y}, respectively, and the surface salinity mask is scaled by α_s.

The default parameter values in Veros are shown in Table 1; the parameterization is used in the two control simulations: CONT1 for the Veros 1° setup and CONT4 for the 4° setup. For the optimization, parameter bounds are necessary to define the search space. The choices are motivated as follows:

- The default c_k and c_ϵ values are taken from Gaspar et al. (1990). Since the two parameters scale $\bar{e}$ production and dissipation, a natural pick for the scale for both parameters is $[0, 1]$. However, MLD becomes noisy at c_k values close to zero; and with no sink for $\bar{e}$, its prognostic equation might become unstable. The minimum values for c_k and c_ϵ should therefore be larger than zero. We pick 0.05 as the minimum bound for both.
- The default value for α_{tke} in Gaspar et al. (1990) is 1, but was increased to 30 when implemented in an ocean model with realistic geometry for the first time to ensure numerical stability (Blanke & Delecluse, 1993). For the bounds, we adapt the range used in Williamson et al. (2017): [0.5, 30].
- The Prandtl number is set to 10 in the ocean interior, i.e. $min_{K_h} = 0.1 min_{K_m}$. The default $min_{K_m} = 2{\cdot}10^{-4}$ is picked based on the observed diffusivity: $K_h \approx 1.7{\cdot}10^{-5}$ m^2 s^{-1} (Ledwell et al., 1998). The lowest values of K_h in microstructure profiles are typically on the order of 10^{-7} m^2 s^{-1}. At the higher end of the range, min_{K_h} imposes a constant background diffusivity below the mixed layer.
- In Eden and Greatbatch (2008), the default eke_c_k and eke_c_ϵ values are 1 and 0.1, respectively. Veros inherits the alternative values picked for the Python Ocean Model (PyOM2.0, Häfner et al., 2017): 0.5 and 0.4. Similarly to the TKE scheme, the two parameters scale the production and dissipation of the eddy kinetic energy, but in the case of eke_c_k, the default value lays on the boundary of [0,1]. We thus opt to vary eke_c_k and eke_c_ϵ on a logarithmic scale spanning four orders of magnitude: $[10^{-2}, 10]$. The wide extent of the scale may lead to unphysical solutions, but ensures a higher likelihood that the optimal solution lies within the parameter bounds.
- A similar line of thought is applied to the choice of parameter bounds for c_{τ_x}, c_{τ_y} and α_s. The maximum wind stress in Veros with the default forcing is 0.45 Nm^{-2}. When $c_{\tau_x} = c_{\tau_y} = 0.6$, this value is reduced to 0.27 Nm^{-2}. Conversely, when $c_{\tau_x} = c_{\tau_y} = 1.4$, the maximum wind stress becomes 0.63 Nm^{-2}. The range of surface salinity spans [39.5 psu, 25.3 psu]. For $\alpha_s = 0.9$, this range changes to [35.5 psu, 22.8 psu]. For $\alpha_s = 1.1$, the range is [43.4 psu, 27.9 psu].

The forcing parameter bounds are reexamined in the OBS_9D experiment and reduced, as discussed in section 3.3.

3.1 Mixed layer depth definition and climatology

The depth of the mixed layer does not have a unique definition (de Boyer Montégut et al., 2004). Density profile observations outnumber diffusivity profiles by orders of magnitude, therefore most MLD definitions use a density threshold criterion of the form:

$$\rho(z) = \rho(z_{\mathrm{ref}}) + \delta\rho, \tag{7}$$

where ρ is the potential density, z is the vertical coordinate, z_{ref} is the reference depth and $\delta\rho$ is a fixed potential density difference. MLD is thus defined as the maximum depth for which the seawater potential density is $\delta\rho$ larger than at the reference depth.

We pick the latest dataset from de Boyer Montégut (2022) as the target MLD climatology (CLIM). The dataset uses a fixed potential density threshold of $\delta\rho = 0.03$ kg m^{-3} and $z_{\mathrm{ref}} = 10$ m for computing the MLD. The climatology is based on shipboard observations, drifter and buoy data. The climatology and the CONT1 MLD bias are shown in Figure 2. MLD in Veros is generally too shallow in the tropics and mid-latitudes, with

Table 1. The default values of the optimized parameters and their bounds. Note that in OBS_9D, min_{K_m} is used instead of $\log_{10}(min_{K_m})$, and its bounds are reduced to [10^{-6}, 10^{-4}].

parameter	default value	bounds
c_k	0.1	[0.05, 1.0]
c_ϵ	0.7	[0.05, 1.0]
α_{tke}	30.0	[0.5, 50]
$\log_{10}(min_{K_m}$ [m^2 s^{-1}])	$\log_{10}(2 \cdot 10^{-4})$	[-7.0, -3.5]
$\log_{10}(eke\_c_k)$	$\log_{10}(0.4)$	[-2.0, 1.0]
$\log_{10}(eke\_c_\epsilon)$	$\log_{10}(0.5)$	[-2.0, 1.0]
c_{τ_x}	1.0	[0.6, 1.4]
c_{τ_y}	1.0	[0.6, 1.4]
α_s	1.0	[0.9, 1.1]

the exception of the Intertropical Convergence Zone (ITCZ) region and the northern Indian Ocean. At high latitudes, the mixed layer is biased deep. The magnitude and direction of the biases in Veros is comparable to the coarse models in the Ocean Model Intercomparison Project (OMIP, Griffies et al., 2016; Treguier et al., 2023).

Treguier et al. (2023) find that MLD biases as large as 100 m can arise from using model reference depth z_{ref} different from the one used to compute the MLD climatology. We therefore set up the vertical grid in Veros so the z_{ref} from the climatology (10 m)

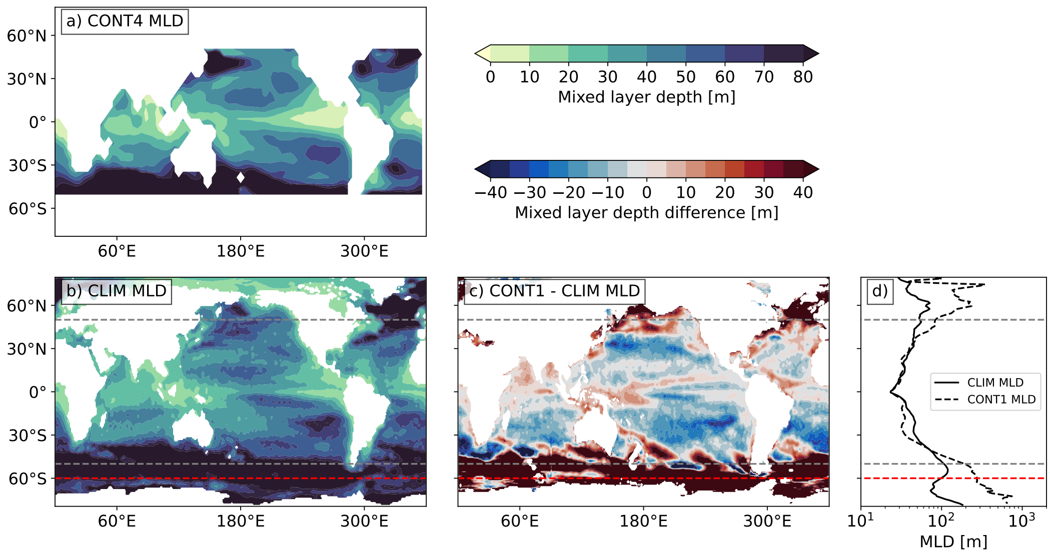

Figure 2. a) CONT4 MLD field used as the optimization target in TWIN; b) the annual MLD climatology used as the optimization target in OBS_TKE and OBS_9D; and c) CONT1 MLD bias. In d), the mean zonal MLD is compared between observations (solid black line) and Veros (dashed black line). The gray dashed lines indicate the meridional extent of TWIN and OBS_TKE. The red dashed line shows the extended meridional bound of the OBS_9D experiment.

matches the model reference depth closely. In Veros $1^\circ \times 1^\circ$, $z_{\mathrm{ref}} = 10.7$ m; in Veros $4^\circ \times 4^\circ$, $z_{\mathrm{ref}} = 11.2$ m. MLD is computed at every time step. The depth at which the potential density criterion in Eq. (7) is fulfilled is found by linearly interpolating between the vertical layers:

$$\mathrm{MLD} = \frac{\rho(\mathrm{MLD}) - \rho(z_b)}{\rho(z_a) - \rho(z_b)} \cdot (z_a - z_b) + z_b, \tag{8}$$

where z_a is the depth layer above MLD, and z_b is the depth layer below MLD. The potential density $\rho(\mathrm{MLD}) = \rho(z_{\mathrm{ref}}) + 0.03$ kg m^{-3}.

3.2 TKE optimization: TWIN and OBS_TKE

The TKE scheme is tuned in two optimization studies: the TWIN experiment, where the target is the CONT4 MLD, and the OBS_TKE experiment, where the target is the MLD climatology. The OBS_9D routine is described in section 3.3. Table 2 outlines the details of all optimization setups.

The goal of the TWIN experiment is two-fold: assess the sensitivity of the modeled MLD to the TKE parameters and determine how well VerOpt can identify a set of known parameters. When ocean simulations are compared to climatology, a degree of bias is expected even in a perfect model due to observational uncertainty. In practice, the model is never perfect, and structural limitations independent of the optimized parameterization are another source of error. The TWIN experiment enables us to assess the performance of the optimizer independently of these two factors.

The objective function in the TWIN experiment, $f_{\mathrm{TWIN}}(\mathbf{x})$, is the natural logarithm of the root mean squared relative error of the MLD map simulated with a parameter set $\mathbf{x}$ (MLD^{sim}) relative to the target MLD, MLD^{target}:

$$rmse = \sqrt{\frac{1}{NM}\sum_{i=0}^{N}\sum_{j=0}^{M}\left(\frac{\mathrm{MLD}_{i,j}^{sim} - \mathrm{MLD}_{i,j}^{target}}{\mathrm{MLD}_{i,j}^{target}}\right)^2}. \tag{9}$$

The dimensions of the map are $N \times M$, with i and j symbolizing the zonal and meridional indices, respectively. The target maps for both experiments are shown in Figure 2. MLD is by definition larger than 10 m, so division by zero is not a risk.

The space $\mathcal{X}_{\mathrm{TWIN}} = \{c_k, c_\epsilon, \alpha_{tke}, min_{\kappa_M}\}$ includes the full TKE parameterization. The experiment is conducted using the 4° Veros setup. The hyper-parameter β of the acquisition function (Eq. (5)) is set to 0.8. The choice is rooted in the tendency of the optimizer to pick boundary parameter values in the test experiments. The optimization

Table 2. The setup of the optimization studies.

	TWIN	OBS_TKE	OBS_9D
target MLD	simulated MLD	MLD climatology	MLD climatology
horizontal resolution	4°	1°	1°
latitude bounds	[52°S, 52°N]	[49.5°S, 49.5°N]	[59.5°S, 49.5°N]
parameterizations	TKE	TKE (c_k and c_ϵ)	TKE, EKE, forcing
total simulations	180	40	624
initial points n_{init}	30	10	48
Bayes points n_{Bayes}	150	30	576
parallel simulations n_{evals}	5	2	32
β	0.8	3	0.8

starts with 6 rounds of initialization where 30 n_{init} points are evaluated, followed by 30 rounds of optimization, with a total of 150 n_{Bayes} evaluated points.

In the OBS_TKE experiment, the 1°×1° Veros setup is used. The goal of the optimization is to minimize the model MLD bias relative to climatology. The parameter space is reduced to two dimensions: $\mathcal{X}_{\text{OBS\_TKE}} = \{c_k, c_\epsilon\}$ based on the results of the TWIN experiment further discussed in section 4.1. The reduction of dimensionality is motivated by computational limitations, as a single 1° 30-year simulation takes about 80 times longer to complete than a 4° run of the same length. A reduced parameter space requires fewer evaluated points to explore effectively. (Note that this limitation is lifted in the OBS_9D experiment, see section 3.4).

In contrast to the TWIN experiment where the exact solution is known, we expect the objective function landscape may be more complex in OBS_TKE. The objective function formulation is therefore modified in the OBS_TKE experiment to reduce its sensitivity to small fluctuations in error. The natural logarithm is dropped, making $f_{\text{OBS\_TKE}}(\mathbf{x}) = rmse$ from Eq. (9). The total of $n = 40$ points are evaluated, including 10 n_{init} points in 5 rounds of initialization and 30 n_{Bayes} points in 15 rounds of optimization.

3.3 Extended parameter space optimization: OBS_9D

The results of the OBS_TKE optimization reveal that here, the vertical mixing scheme parameters in isolation cannot be tuned to reduce MLD biases (see section 4.2). Motivated by this, we perform an additional experiment with the search space extended to nine model parameters: the four TKE parameters, eke_c_k and eke_c_ϵ from the mesoscale eddy closure, and the three parameters scaling the surface wind stress and salinity masks. In summary, we have:

$$\mathcal{X}_{\text{OBS\_9D}} = \{c_k, c_\epsilon, \alpha_{tke}, min_{K_m}, eke\_c_k, eke\_c_\epsilon, c_{\tau_x}, c_{\tau_y}, \alpha_s\}.$$

Just as in the case of OBS_TKE, the goal of OBS_9D is to minimize Veros MLD biases.

The increase in problem complexity requires a higher number of objective function evaluations. The optimal number of simulations is highly dependent on the optimization problem, although in Bayesian optimization literature, 400-600 evaluations are often picked for moderate- and high-dimensional parameter spaces (e.g., Hvarfner et al., 2024). For this extended experiment, we make use of the Lumi supercomputer, allowing us to run Veros more efficiently than in TWIN and OBS_TKE (see section 3.4). Because of this, we settle on 2 rounds of initialization and 18 rounds of optimization with $n_{evals} = 32$ parallel Veros simulations. In total, 624 simulations are initiated in the OBS_9D experiment.

The first round of initialization is compiled from 16 test runs performed as we set up Veros and VerOpt on Lumi. All following batches consist of 32 points each. Out of 624 simulations, 375 complete at 60 years and the rest (40%) diverges. It is unclear whether specific parameter combinations make the simulations unstable, so we do not introduce any punishment for the diverging solutions. As a result, on average 19 new objective function evaluations per optimization step are used to update the surrogate model. This leaves significant room for improvement in terms of optimization efficiency, and will be a subject of further study.

Initially, the objective function $f_{\text{OBS\_9D}}(\mathbf{x})$ is the $rmse$ between the simulated and climatological MLD fields, identical to the OBS_TKE experiment. However, after the eighth batch, we notice that some solutions with realistic MLD fields show significant biases in general circulation patterns. This leads to two alterations in the experimental setup: 1) the reduction of the forcing field parameter bounds and 2) the inclusion of additional error metrics.

The first change is motivated by our initial guess that allowing wind stress to vary by ±40% significantly impacts the surface ocean currents and leads to unphysical solutions. Further analysis in section 4.3 reveals that instead, it is the large eke_c_k values causing the deep circulation to shut down in most cases. While the eke_c_k bounds are not altered, introducing additional error metrics in the objective function is enough to deter the optimizer from picking the large eke_c_k values. The bounds of c_{τ_x} and c_{τ_y} are reduced from [0.6,1.4] to [0.85,1.15], and the bounds of α_s from [0.9,1.1] to [0.95,1.05]. Additionally, we catch a mistake where the bounds of min_{K_m} are set to [10^{-6},10^{-5}] instead of [10^{-6},10^{-4}]. These are changed as well.

With the second change, four additional error metrics are added to the objective function: boundary punishment, the Antarcitc Circumpolar Current (ACC) strength, the Atlantic Meridional Overturning Circulation (AMOC) strength and the bias in the depth of the 18°C isotherm.

The boundary punishment term increases exponentially from 0 to 1 as the parameter vector $\mathbf{x}$ approaches the limits of the parameter space. This term is introduced to counteract the known tendency of Bayesian optimizers to over-sample at boundaries (see discussion). The exact formulation of the term is described in Appendix B1.

The difference between the simulated and observed strengths of the AMOC and the ACC is used to compute the AMOC$_{err}$ and the ACC$_{err}$ terms, respectively. The target values and their uncertainties are: $\mathrm{AMOC}_{target} \pm \sigma_{\mathrm{AMOC}} = 17.6 \pm 4$ Sv and $\mathrm{ACC}_{target} \pm \sigma_{\mathrm{ACC}} = 155 \pm 19$ Sv (Cunningham et al., 2003; Donohue et al., 2016; Fu et al., 2020). If the modeled transport strength falls within the uncertainty bounds, the error term is set to zero. Outside of the uncertainty bounds, it grows exponentially until it reaches a maximum of 1. Further details are described in Appendix B2.

The thermocline depth is constrained by comparing the simulated and observed depths of the 18°C isotherm with the 18C$_{err}$ term. The observed profile is based on temperature climatology from World Ocean Atlas 2023 (WOA23, Reagan et al., 2024). If the modeled zonal average of the 18°C depth falls within the observational uncertainty bounds, the term is set to zero. Outside of the bounds, 18C$_{err}$ is computed using the absolute difference between the modeled and observed 18°C depth. The exact formulation is reported in Appendix B3. We find that the 18C$_{err}$ term is generally an order of magnitude smaller than the others, thus it is multiplied by 3 in $f_{\mathrm{OBS\_9D}}(\mathbf{x})$. Finally, the MLD$_{err}$ term is equal to $rmse$ in Eq. (9). The new objective function is defined as:

$$f_{\mathrm{OBS\_9D}}(\mathbf{x}) = \mathrm{MLD}_{err} + 3 \cdot 18\mathrm{C}_{err} + \mathrm{ACC}_{err} + \mathrm{AMOC}_{err} + \text{boundary punishment}. \quad (10)$$

The default VerOpt setting of the hyper-parameter $\beta = 3$ used in OBS_TKE makes the optimizer heavily prioritize exploration over exploitation. The results of the TWIN and OBS_TKE experiments indicate that the objective function maximum is likely to be a surface spanning large regions of the parameter space. We thus reduce β to 0.8 in OBS_9D, focusing the search in the regions where the mean of the surrogate model is minimized. After the fifteenth batch, we notice that the spread of the objective function values in the OBS_9D experiment remains relatively constant. To test whether the acquisition function parameters can be altered such that the optimizer prioritizes exploitation even further, we reduce β to 0.2. In OBS_9D, 32 points are selected for evaluation in parallel, indicating that the exponential distance punishment may dominate the acquisition function landscape. We therefore reduce the radius and magnitude of the exponent in the $p(\mathbf{x})$ term to $\alpha = 0.05$ and $\omega = 0.3$, respectively.

3.4 Performance and computational resources

Veros can run both fully on CPU, fully on GPU, and a mixture. We conducted the 4° TWIN experiment on the Danish Center for Climate Computing (DC^3) CPU cluster *Aegir*, comprised of Intel Xeon E5-2650v4 2.2GHz CPUs, running each simulation with 24 threads. The 1° OBS_TKE experiment was run on our local workstation with mixed calculations on commodity NVIDIA RTX 3080 GPUs and AMD Threadripper Ryzen 1950X 16-Core CPUs (*Threadripper*). The OBS_9D experiment has been conducted on the Lumi-G supercomputer, fully on AMD MI250X GPUs (*Lumi*).

On a single Lumi MI250X GPU, we can compute about 75 4° model years per hour, and 1.2 1° model years per hour, calculated in full 64-bit floating point precision. The average energy consumption for the GPU while running a 1° calculation was 196W, i.e. 163Wh per model year. The float32/float64 mixed GPU/CPU-calculations on Threadripper computes at a similar speed, and uses about 300W on the GPU and roughly 150W on CPU, i.e. $\sim$ 375Wh per model year. The fully-CPU calculations on Aegir were benchmarked for the same 1° setup by Häfner et al. (2021) for 0.115 model years using on DC^3 Aegir, completing in 3297s and consuming 320Wh (Tables 1 and A1 in Häfner et al., 2021), which corresponds to 8 hours and 2782Wh per 1° model year. Thus the same calculation on a single MI250X GPU was not only nearly 10 times as fast as on the CPUs, but also consume 17 times less energy when running fully on Lumi's MI250X GPUs.

The 4° experiments were run on the same system at DC^3, using a single CPU core per run, at 2.8 minutes per model year. Initially we ran the 1° experiments on the same system, using 16 CPU cores per run, resulting in 11 hours per model year. The move to GPUs and reduction to 50 minutes per 1° model year made experimentation much more flexible. As the energy consumption for performing long term climate simulations is substantial – especially when performing many runs to optimize parameters – the final energy reduction by a factor 17 makes an enormous difference as to which kinds of experiments can be justified. Due to the relatively coarse resolution of the 1°, we ran the experiments on a single GPU per Veros-simulation. However, Häfner et al. (2021) showed that Veros scales well up to at least 16 GPUs for 0.1° simulations, making high-resolution Veros runs feasible on multi-GPU based systems such as Lumi-G.

4 Optimization results

4.1 TWIN experiment

After 36 optimization steps, the best simulation produces $rmse \approx 1\%$. The parameter values from this simulation, $\boldsymbol{\chi}^*_{\mathrm{TWIN}}$, are shown in Table 3 together with percentage difference relative to the target values. While Ri_c is not tuned directly, it turns out to have the strongest influence on the objective function, thus we report it in Table 3 alongside the other parameters.

Table 3. VerOpt best estimates for the TKE parameterization in the TWIN experiment.

parameter	target value	best TWIN estimate	% difference
c_k	0.1	0.12	20
c_ϵ	0.7	0.87	24
α_{tke}	30.0	36	20
min_{κ_M} [$\mathrm{m^2\ s^{-1}}$]	$2 \cdot 10^{-4}$	$1.6 \cdot 10^{-4}$	20
Ri_c	0.241	0.236	2

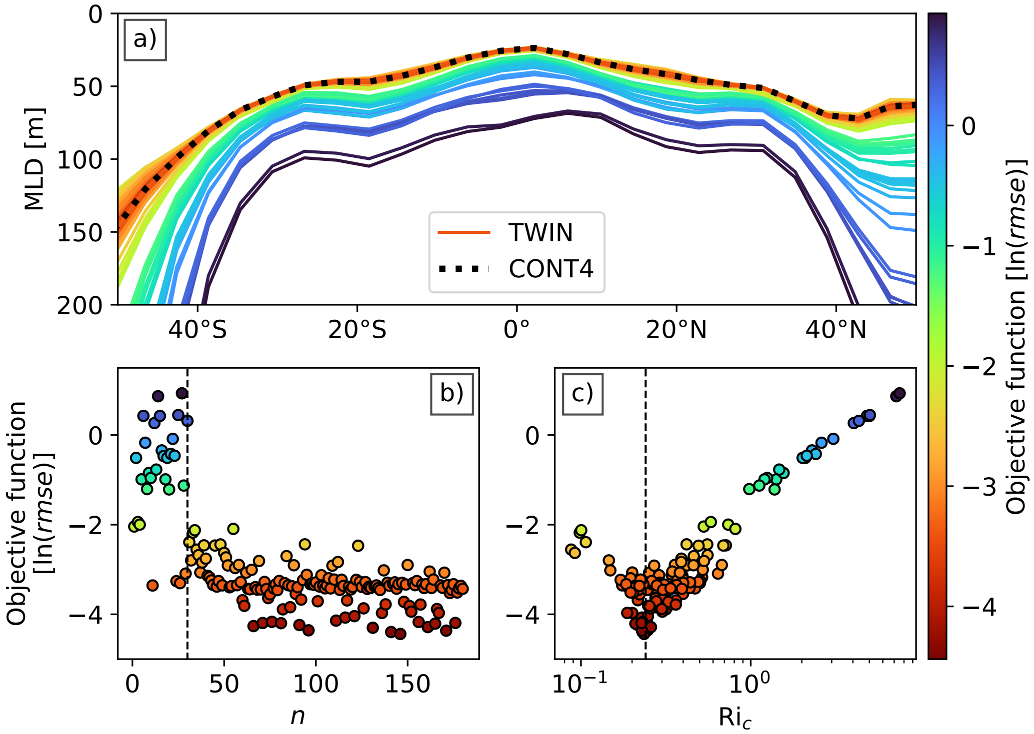

Figure 3. TWIN optimization results: a) zonal MLD averages in the 180 experimental runs and in CONT4; b) objective function progress after n points; c) $\ln(rmse)$ as a function of the critical Richardson number Ri_c. In b), the vertical dashed line indicates the separation between initial and Bayes points. In c), the line indicates the target $Ri_c \approx 0.24$.

In Figure 3c), $\ln(rmse)$ is plotted as a function of the critical Richardson number Ri_c. The global MLD is tightly correlated with Ri_c, as seen in Fig. 3a). Increasing the Ri_c has the effect of deepening the global MLD. The deepening is more pronounced at mid-latitudes, where the stratification is weaker. In Fig. 3b), the objective function values are plotted against the number of evaluated points n. The difference between randomly chosen points and the points suggested by VerOpt is evident.

The optimization results indicate that large regions of the TKE parameter space simulate similar MLD distributions. Moreover, since Ri_c has the highest influence on the MLD, and since it is also a function of $c_k c_\epsilon^{-1}$, local $rmse$ minima will occur for any parameterization where the $c_k c_\epsilon^{-1}$ ratio is the same as default. The relatively large parameter errors in Table 3 are a reflection of this: while c_k and c_ϵ are 20% and 24% larger than the default values, the ratio - and therefore also Ri_c - is only 2% off from target.

The TWIN experiment demonstrates the robustness of the tool. For the price of only 180 objective function evaluations, the optimizer is able to identify the most influential parameters and find a parameter set that simulates the target MLD to 1% accuracy. This confirms that if the optimal parameter values are within the tested bounds, the optimizer can locate the region of the parameter space that best explains the target data. If the bias relative to the target data cannot be minimized beyond a certain threshold, it likely originates from the combination of structural limitations of the model and observational uncertainty.

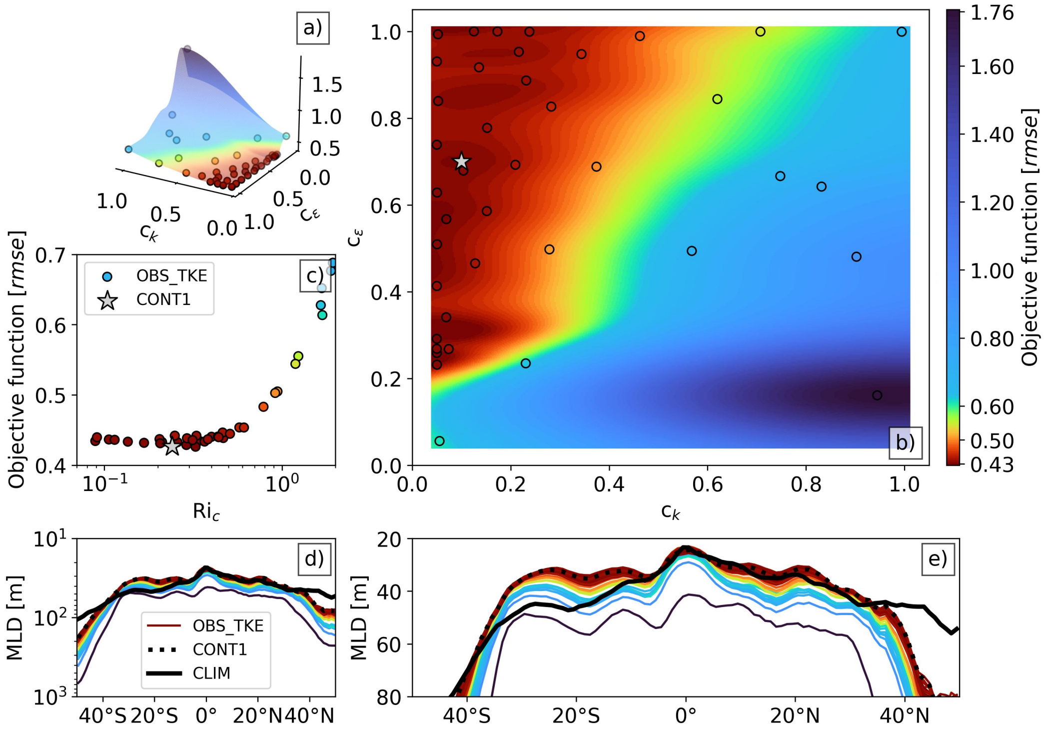

Figure 4. OBS_TKE optimization results: a) 3D and b) 2D visualization of the objective function surrogate model mean $\hat{\mu}$ at the final optimization round and the 40 evaluated objective function coordinates; c) *rmse* as a function of the critical Richardson number Ri_c; d) zonal average MLD in the 40 simulations, CONT1 run and observations; and e) zonal average MLD as in d) with a reduced y-axis scale. Panel d) is included to show the full extent of the MLD biases at higher latitudes; note the logarithmic scale.

4.2 OBS_TKE experiment

In the OBS_TKE experiment, we focus on the two TKE parameters with the most influence over the global MLD in the TWIN experiment: c_k and c_ϵ. Figure 4 shows the overview of the optimization results. The minimum bias in the OBS_TKE experiment is 43%, identical to CONT1. The comparison of the observed and modeled zonal MLD profiles in Fig. 4e) reveals that no set of evaluated TKE parameters brings the global MLD distribution closer to observations. The climatological MLD is asymmetrical about the equator, which is not reflected in any of the models.

In the run with the lowest error, $Ri_c = 0.33$. When the critical Richardson number is less than unity, the MLD bias is relatively insensitive to it. The *rmse* in those runs is 44.4±0.2%. For $Ri_c > 1$, MLD becomes more sensitive to the parameter values, with MLD bias varying between 54% for $Ri_c = 1.2$ and 177% for $Ri_c = 9.9$. The critical Richardson number shifts the global MLD downwards as it increases, similarly to the 4° Veros setup (Fig. 3). The *rmse* is lowest in the upper left quadrant of the parameter space (Fig. 4a).

4.3 OBS_9D experiment

The OBS_9D optimization sequence is visualized in Figure 5 with the objective function values, individual contributions of the error sources and the parameter values plotted over the 20 batches. The alterations to the OBS_9D setup are marked on each panel with vertical gray dotted lines. Note that prior to the eighth batch, MLD$_{err}$ is the objective function. The introduction of the additional error metrics is effective: with the surrogate model conditioned on the points with unrealistic AMOC and ACC, the optimizer successfully samples the region of the parameter space for which ACC$_{err}$ and AMOC$_{err}$ are minimized.

On the other hand, 18C$_{err}$ does not improve significantly after its introduction. We find this is due to a bug in the computation of the error, where 18C$_{err}$ $= 0$ when the depth of the 18°C isotherm is biased shallow. We discuss the possible consequences of this bug on the optimization sequence in Appendix B3. Fortunately, we find that the corrected 18C$_{err}$ values and the objective function values correlate: only the unrealistic solutions correspond to high 18C$_{err}$. The corrected objective function is thus shifted towards higher values, but its distribution is not significantly altered (Fig. B1).

The parameter value distributions suggest which of them have the highest impact on the objective function. Just as in TWIN and OBS_TKE, the parameter c_k and the ratio $c_k c_\epsilon^{-1}$ retain their high influence over the global MLD biases. A well-defined peak can also be seen in the distribution of eke_c_k. Aside from these three, the optimizer shows weak preference for low c_{τ_x}, α_{tke} and eke_c_ϵ values. The objective function appears insensitive to α_s and c_{τ_y}. Note that because of the disproportionate boundary punishment of the lower range of the min_{K_m} values, its distribution does not necessarily reflect the sensitivity of model errors to min_{K_m} (see discussion in Appendix B1).

The impact of the model parameters on the objective function is further explored in Figure 6. The 2D slices of $c_k - c_\epsilon$ and $c_k -$eke_c_k parameter spaces show their dominant impact on the objective function landscape. We also see resemblance to the $c_k - c_\epsilon$ space in OBS_TKE (Fig. 4b), despite the much increased problem complexity of OBS_9D.

The location of the objective function minimum along the eke_c_k parameter is strongly correlated with the corresponding eke_c_ϵ value, as seen in the eke_$c_k -$ eke_c_ϵ parameter space. The flattening of isopycnals due to mesoscale eddy and isoneutral mixing coefficients is another major process that impacts the global MLD. The EKE parameters govern the eddy kinetic energy production and dissipation. The results indicate that sets of eke_c_k and eke_c_ϵ combinations can produce similar, optimal distributions of K_{gm} and K_{iso}.

The model critically fails either when eke_c_k is larger than 0.5 and unbalanced by high dissipation rate eke_c_ϵ, and when $Ri_c > 1.1$. The former results in the weakening of the ACC transport and the shutdown of the AMOC, while the latter deepens the MLD globally. Contrary to our initial assumption, good solutions exist even when zonal wind stress is reduced by 40% or increased by about 30%. In most simulations, the AMOC and ACC strengths fall within or close to the observational uncertainty bounds. The few simulations for which AMOC and/or ACC are too weak correspond to the high eke_c_k values.

The ACC strength is correlated with c_{τ_x}, with the exception of the simulations where the eke_c_k values are the largest. The impact of zonal wind stress on the ACC is debated in literature (e.g., Jochum & Eden, 2015). Numerous studies indicate that there exists a theoretical limit where the wind stress changes over the Southern Ocean impact the mesoscale eddy-induced circulation only, and do not result in weakening or strengthening of the ACC (the so-called eddy saturation), although eddy-resolving models do not always show this (see discussion in Poulsen et al., 2018). Our results indicate that eddy saturation is not present in the model within the bounds of the tested wind stress limit.

Nearly all parameter vectors suggested by VerOpt lead to MLD bias reduction. Compared to the default $rmse = 0.63$, the three best solutions are improved by 23% with

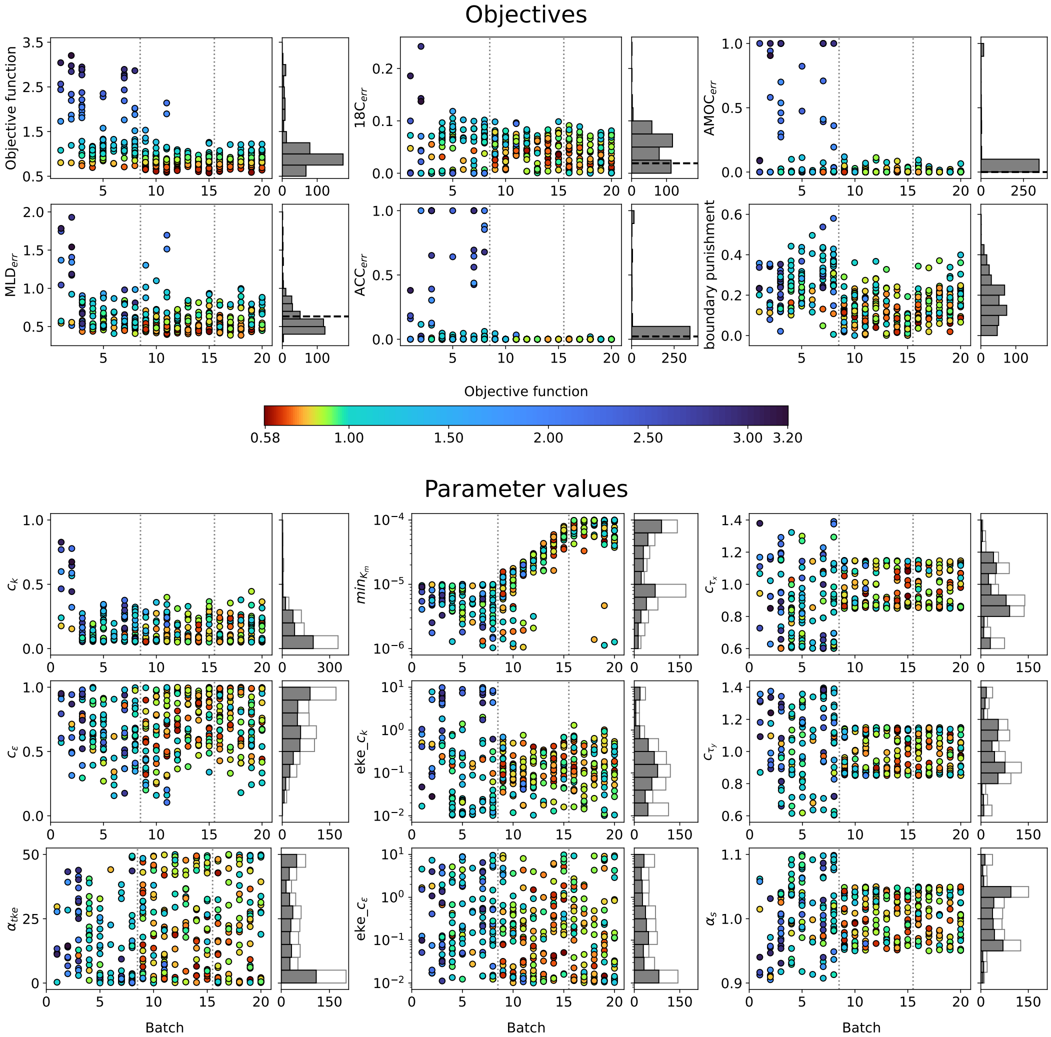

Figure 5. OBS_9D optimization sequence. The six top panels show the objective function values over the two initialization and 18 optimization rounds, as well as the individual error contributions from the MLD *rmse* error (MLD_{err}), the depth of the 18°C isotherm error ($18C_{err}$), the deviation from the observed AMOC and ACC strength ($AMOC_{err}$ and ACC_{err}), and the boundary punishment term. The histograms show the spread of the values over the entire optimization sequence. The CONT1 errors are marked on the histograms with black dashed lines. The nine bottom panels show the parameter values of the 375 completed Veros simulations. The gray bars on the corresponding histograms show their distributions, and the white bars show the distributions from the remaining 249 diverged solutions. The dotted lines on each panel indicate the instances at which changes were made to the OBS_9D experiment: after the 8th batch, the objective function was changed to include additional error metrics and the bounds of c_{τ_x}, c_{τ_y}, α_s and $\log(min_{K_m})$ were updated. After the 15th batch, the acquisition function hyper-parameters were updated.

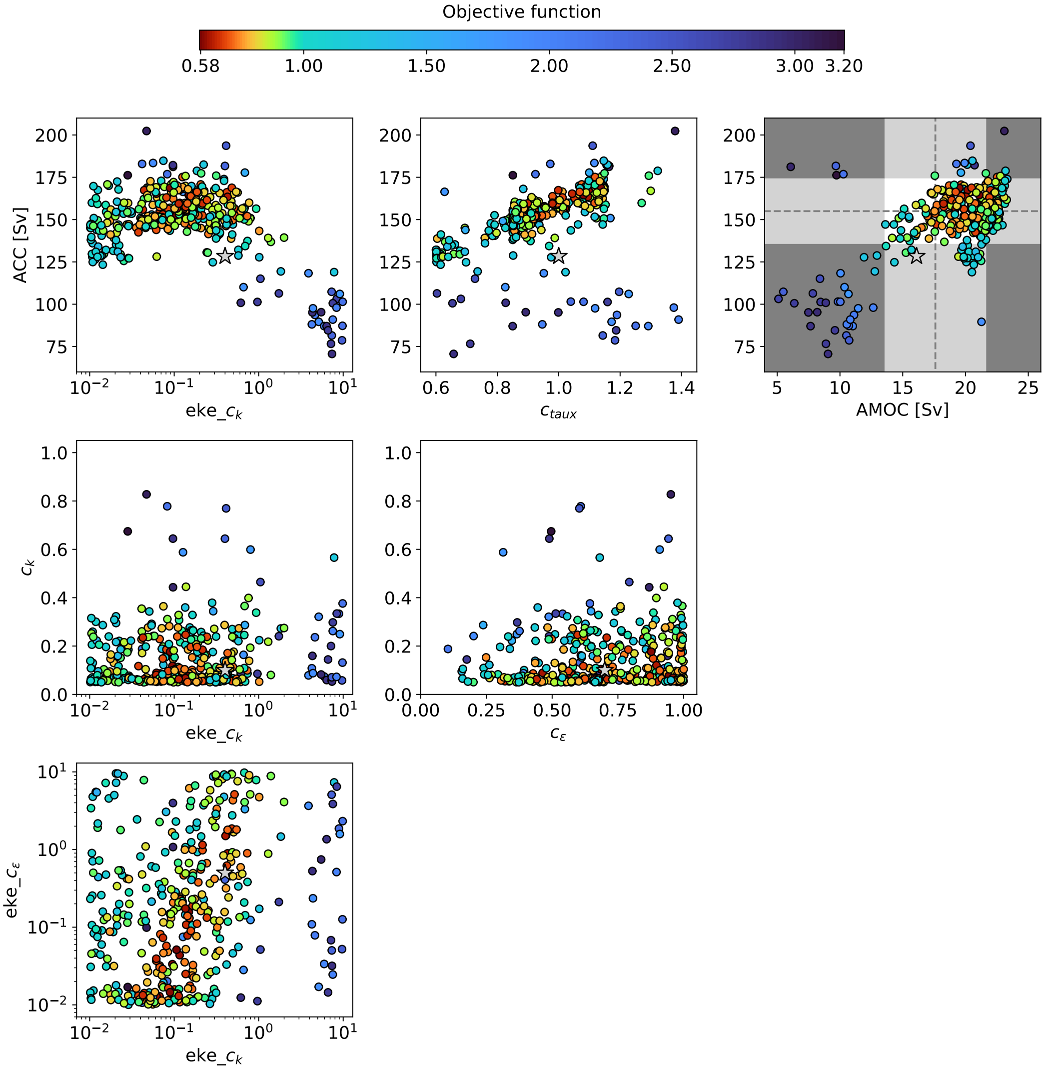

Figure 6. Overview of the OBS_9D parameters with the highest impact on the objective function value, as well as their influence over the circulation patterns in the model (note the difference in the x-axis parameters in the middle panels). CONT1 values are indicated with a gray star. The upper right panel shows the ACC and AMOC strength across the 375 OBS_9D simulations. The light gray regions span over $\mathrm{ACC}_{target} \pm \sigma_{\mathrm{ACC}}$ and $\mathrm{AMOC}_{target} \pm \sigma_{\mathrm{AMOC}}$. For all simulations within the white rectangle, $\mathrm{AMOC}_{err} = \mathrm{ACC}_{err} = 0$. In the dark gray region, both $\mathrm{AMOC}_{err} > 0$ and $\mathrm{ACC}_{err} > 0$.

$rmse < 0.4$. The parameter values of these simulations are listed below:

$$\begin{aligned} \mathcal{X}_{\text{OBS\_9D}} &= \{c_k, & c_\epsilon, && \alpha_{tke}, & min_{K_m}, & eke\_c_k, & eke\_c_\epsilon, & c_{\tau_x}, & c_{\tau_y}, & \alpha_s\} \\ \mathbf{x}_{\text{CONT1}} &= [\,0.1, & 0.7, && 30, & 2.0\cdot 10^{-5}, & 0.4, & 0.5, & 1, & 1, & 1] \\ \boldsymbol{\chi}_1^* &= [\,0.05, & 1.0, && 49, & 9.7\cdot 10^{-5}, & 0.07, & 0.05, & 0.86, & 0.90, & 1.05] \\ \boldsymbol{\chi}_2^* &= [\,0.06, & 0.5, && 1.4, & 7.5\cdot 10^{-5}, & 0.05, & 0.01, & 0.85, & 1.14, & 1.05] \\ \boldsymbol{\chi}_3^* &= [\,0.09, & 0.5, && 1.4, & 1.6\cdot 10^{-5}, & 0.06, & 0.01, & 0.86, & 1.12, & 1.05], \end{aligned} \tag{11}$$

where $\mathbf{x}_{\text{CONT1}}$ is the default parameter vector.

What are the similarities between the models for which the MLD bias is minimized? In order to answer this question, we analyze an ensemble of 39 solutions where $rmse < 0.45$ (OPT_MLD). Figure 7 summarizes our findings. Panel a) shows that the most significant bias reduction occurs at high latitudes, especially in the Southern Ocean (SO). Similarly to OBS_TKE, the $rmse$ dramatically increases for Ri_c values above unity. Relative MLD biases and mean zonal K_{gm} profiles in CONT1 and the OPT_MLD ensemble mean ($\overline{\text{OPT\_MLD}}$) are compared in panels e) and f), and panel g) compares the two. The absolute MLD bias difference is defined as $|\overline{\text{OPT\_MLD}}-\text{CLIM}|-|\text{CONT1}-\text{CLIM}|$, and in the K_{gm} ratio, the $\overline{\text{OPT\_MLD}}$ K_{gm} is divided by the K_{gm} in CONT1.

In most of the OPT_MLD solutions, $c_{\tau_x} < 0.9$, $c_k < 0.1$, and the value of eke_c_k peaks at around 0.1. The zonally averaged K_{gm} is about halved compared to the control run. The mid-latitude mixed layer is deeper in the $\overline{\text{OPT\_MLD}}$ ensemble mean due to the reduction of K_{gm} and K_{iso}, which generally steepens the isopycnals. In panel c), it is evident that the equatorial thermocline depth is increased in OPT_MLD compared to CONT1. In the SO, the ensemble mean mixed layer is shallower, improving the biases from CONT1 by up to 340 m. The zonally averaged SO isopycnals are flatter in the OPT_MLD runs, indicating that the weaker zonal wind stress is the dominant factor in improving the simulated MLD in this region.

5 Discussion

VerOpt is used to identify the set of parameter values that minimize Veros MLD biases. In the TWIN experiment, the optimization target is MLD simulated using the Veros 4° setup. The lowest $rmse$ of 1% is achieved with TKE parameters that deviate from the target parameters by about 20-25% (Table 3). The target of the OBS_TKE experiment with the reduced parameter space $\mathcal{X}_{\text{OBS\_TKE}} = \{c_k, c_\epsilon\}$ is the MLD climatology (de Boyer Montégut, 2022). We show that tuning the TKE scheme alone is not sufficient to rectify the MLD biases in the model. In the OBS_9D optimization, where the parameter space is expanded to include the mesoscale eddy closure and the surface forcing fields, the MLD biases are reduced by up to 23%. We diagnose an ensemble of 39 simulations where the MLD $rmse < 0.45$. The K_{gm} coefficient in these runs is approximately halved and the zonal wind stress is 10% weaker compared to the control run with default parameterization.

Is the modeled mixed layer depth improved for the "right" reasons? Although the parameter values found in the OBS_9D optimization significantly improve the simulated MLD in Veros, substantial biases remain (Fig. 7). Due to the coarse 1° grid spacing, the dynamics of the Kuroshio current and the Gulf Stream are not resolved in the model, likely leading to the persistent MLD biases in the Northwest Atlantic and Northwest Pacific seen both in CONT1 and in $\overline{\text{OPT\_MLD}}$. In the current experiments, the Veros setup does not include dynamic sea ice, which may introduce surface salinity biases in the regions of deep convection that propagate to lower latitudes. Numerous vertical mixing processes that have been shown to impact the depth of the mixed layer (e.g., Langmuir turbulence or near-inertial waves; Jochum et al., 2013; Li et al., 2019) are not included in the TKE parameterization in Veros. The 18°C isotherm is too deep around the equa-

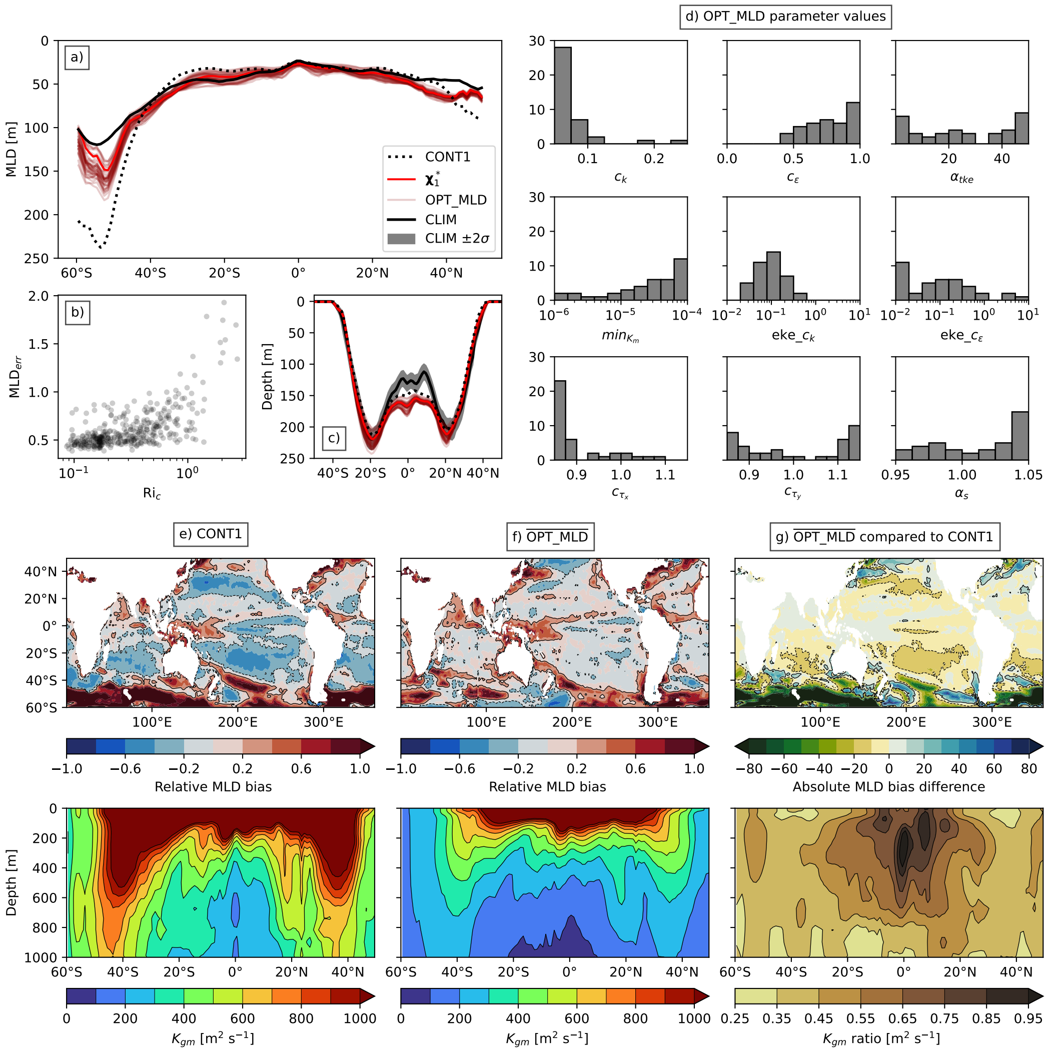

Figure 7. Overview of the OBS_9D runs with MLD$_{err}$ < 0.45 (OPT_MLD): a) zonal average MLD compared to CONT1 and climatology with the best solution (χ_1^*) highlighted; b) MLD$_{err}$ as a function of Ri_c; c) zonally averaged depth of the 18°C isotherm compared to CONT1 and WOA23 observations, with the uncertainty indicated by the gray shading and the best solution (χ_1^*) highlighted; d) distributions of the OPT_MLD parameter values; e) the relative MLD bias and the zonally averaged K_{gm} coefficient in CONT1; f) same as e) but in the mean of the OPT_MLD ensemble; g) the ensemble mean compared to CONT1. In e) and f), relative MLD bias of -0.2 and 0.2 is marked by a black contour. In g), the contours mark -10 m and 10 m, and green indicates the regions where the absolute MLD bias is reduced.

tor across all OPT_MLD simulations, which may be caused by the absence of internal wave mixing in the model (Jochum, 2009). Model calibration that targets steady-state solutions is prone to compensating biases (Hourdin et al., 2017). Therefore, it is impossible to state with certainty that the improved MLD distribution in OPT_MLD indicates the model is a closer representation of the real ocean than CONT1.

We do, however, gain a wealth of information that can be used to further the effort to understand and eliminate MLD biases in OGCMs. Consistent with the studies of the second-moment closure schemes, we confirm that $c_k c_\epsilon^{-1} \propto Ri_c$ is the dominant parameter combination setting the global MLD in the TKE scheme (Burchard & Bolding, 2001). In the present setup, the diffusion coefficients K_{iso} and K_{gm} are equal. The significant impact of eke_c_k on the MLD field, the circulation and the thermocline depth reveals the ratio of K_{iso} and K_{gm} as a potential key target for tuning. OBS_9D optimization also reveals that out of the 39 OPT_MLD simulations, 29 include $c_{\tau_x} < 0.9$. Belmonte Rivas and Stoffelen (2019) compare ERA-Interim zonal wind stress to satellite observations and show that it is too strong by about 10%, consistent with our optimal c_{τ_x} values.

Williamson et al. (2017) propose an alternative method for automated model tuning that also utilizes GP regression, but employs a different sampling method. They search a 24-dimensional parameter space of the NEMO ORCA2 model. Despite differences in methodology and the model used, the results of their study are remarkably similar to ours. The TKE parameters c_k and c_ϵ have the highest influence on the temperature in the upper 300m in ORCA2, alongside the Langmuir cell coefficient and the surface TKE input coefficient that are not included in Veros. The majority of the parameter space tested in Williamson et al. (2017) results in a warm bias in the upper ocean, indicating excessive deepening of the ML. Similarly, we find that increasing the $c_k c_\epsilon^{-1}$ ratio in $1^\circ \times 1^\circ$ Veros leads to global MLD increase, and that MLD is generally the same or deeper compared to CONT1 in all of the OBS_TKE simulations (Fig. 4).

Climate models differ not only in tunable parameters, but also fundamental building blocks such as grid type, geometry, resolution, components, etc. - all of which affect the simulated output. Therefore, parameter tuning is highly model dependent, and an optimal choice of parameter values for one model is not guaranteed to be the best choice for another. However, the similarities between the results in this work and in Williamson et al. (2017) suggest that *some* information about commonly used parameterizations could be transferable between different models. MLD deepening as a result of increasing the $c_k c_\epsilon^{-1}$ ratio in the TKE scheme is expected to occur in models other than Veros, as a higher critical Richardson number necessarily results in enhanced vertical diffusivity.

One major limitation of the optimization experiments is that we tune annual mean mixed layer instead of considering summer and winter biases separately. The choice to consider annual mean MLD originates from the limited storage space. In OBS_9D, storing 375 60-year long simulations at seasonal temporal frequency would be unfeasible. To investigate the impact of this limitation on the optimization results, we run three simulations with the best parameters listed in Eq. (11) with monthly temporal resolution. We find that annual means are dominated by winter MLD biases, especially in the Southern Ocean. Compared to CONT1, winter MLD is improved in most of the SO and some regions in the Northern Atlantic and Pacific, and in Southern mid-latitudes. However, summer mixed layer is biased shallow in both hemispheres, and the biases are intensified in the simulations that use the optimized parameters. Generally, the magnitude of winter MLD biases in OGCMs is much larger than summer biases, which makes it difficult to construct a cost function that targets both simultaneously. Summer MLD biases could therefore be a good target for future calibration experiments.

The model bias diagnostics presented here are focused on global annual mean fields, but regional bias patterns could be subject to further analysis. The 375 OBS_9D simulations provide a basis for investigating how the tested parameter combinations impact model state variables not considered in our work. Alternative error metrics based on eddy kinetic energy biases, vertical diffusivity fields or buoyancy frequency profiles could be formulated and their sensitivity to model parameters tested to guide future calibration studies.

Our work uses a relatively low-dimensional example to demonstrate the utility of Bayesian optimization in climate modeling. The goal of this paper is to introduce VerOpt in its basic form, but many aspects of the optimization could be enhanced. The BO toolkit has grown significantly over the past decades due to its popularity (Wang et al., 2023). The following paragraphs outline the improvements particularly relevant to climate modeling.

Search in higher dimensions. High-dimensional optimization problems suffer the curse of dimensionality. The expensive to evaluate objective functions pose an additional challenge in this regard. If the number of points necessary to construct an accurate GP model in one dimension is n, then in principle for d dimensions we would need n^d points. This is of course in conflict with the desire to minimize the necessary objective function evaluations. In practice, optimizer performance is highly dependent on the shape of the objective function. In the TWIN experiment, the GP model constructed with only 40 input points is able to accurately locate the region in the parameter space where $rmse$ is below 13% (Fig. 3b). Were the objective function highly nonlinear, this would likely require more Veros simulations. BO is used in high-dimensional optimization problems across many disciplines, thus a plethora of methods for coping with the curse of dimensionality has been proposed. Two most commonly used tactics are variable selection and low-dimensional embeddings. The former involves reducing the search space based on the relative influence of parameters on the objective function (Chen et al., 2012). This strategy has been informally applied in this work, as we reduced the search space in the OBS_TKE experiment based on the findings from the TWIN optimization. An alternative to the exclusion of inactive parameters is to optimize a low-dimensional embedding of the high-dimensional parameter space, the so-called *active space* (e.g. Nayebi et al., 2019). For example, the linear combination of c_k and c_ϵ^{-1} could be optimized in the TKE scheme instead of considering the two parameters separately.

Over-sampling at the boundaries. The *boundary issue* is a common challenge in BO (Swersky, 2017), where the optimizer over-samples the edges of the parameter space. In the TWIN and OBS_9D experiments, we attempt to mitigate this tendency by reducing the value of β in the acquisition function to 0.8, but this comes at a cost of potentially under-exploring the regions in the parameter space where the information about the objective function is lacking. The boundary issue is noticeable in the OBS_TKE experiment (Fig. 4b). In OBS_9D, we introduce a term in the objective function to deter the optimizer from sampling boundary values. A more robust strategy to combat this tendency that does not impact the value of the objective function directly has been proposed by Oh et al. (2018). The method, Bayesian Optimization with Cylindrical Kernels (BOCK), transforms the geometry of the search space to increase its volume at the center and reduce it at the boundaries. BOCK has been shown to successfully scale to 500-dimensional optimization problems and is already implemented in BoTorch (Balandat et al., 2020).

The error metric and MLD uncertainty. The work of Williamson et al. (2017) underlines the importance of accounting for observational uncertainty in the tuning process. In this work, we do not incorporate the information about MLD uncertainty in the objective function, as it is not included in the dataset we use. However, MLD in the climatology from de Boyer Montégut (2022) is computed using a constant density threshold of 0.03 kg m^{-3}, as opposed to other products that typically use a variable density threshold to match the $\pm 0.2°$ C temperature criterion (de Boyer Montégut et al., 2004). This is done to make model comparison with observations easier, as nearly all OMIP models use constant density threshold for MLD computation (Treguier et al., 2023). The MLD uncertainty could be incorporated into the objective function by e.g. redefining the error metric. Instead of computing the relative distance between MLD values in the simulated and target maps (Eq. (9)), the distance between MLD *distribution* in each grid cell could be used.

Multiple objectives. Climate model optimization always involves pay-offs between the accuracy of different processes at various geographical locations. Tuning to a single objective can easily return a parameterization that critically misrepresents another aspect of the climate, as was in the case of the OBS_9D experiment where model simulations with realistic MLD fields were accompanied by large biases in the strength of the overturning circulation. In OBS_9D, additional error metrics are combined to form a single objective function, but this approach is limited. For example, the relative magnitude of the different error metrics is not considered. Multi-objective Bayesian optimization accounts for this limitation with specialized acquisition functions. The multi-objective optimization in VerOpt employs the Expected Hypervolume Improvement acquisition function (Emmerich et al., 2006). While conducting the OBS_9D experiment, VerOpt was not yet optimized to efficiently handle more than three targets, but the latest release (v0.6.0) supports multi-objective optimization.

The work presented in this paper only scratches the surface of the potential BO has in automated tuning of climate models. Due to its popularity, it has undergone rapid development in recent years. This has resulted in the emergence of specialized algorithms, which make BO highly adaptable to a wide range of problems. VerOpt makes BO seamlessly applicable to climate modeling. It can work with any model output, but it benefits significantly from Veros Python/JAX infrastructure, which supports GPU computation, enabling faster and more energy efficient simulation.

6 Conclusions

We propose Bayesian optimization with VerOpt as a method for automated parameter tuning in OGCMs. Three optimization experiments demonstrate the process by which VerOpt identifies a set of parameter values that minimize the simulated MLD error relative to a target map. In the idealized TWIN experiment, the best *rmse* is 1%. The OBS_TKE experiment demonstrates that the TKE scheme alone cannot be tuned to reduce MLD biases in Veros. We achieve this instead with the parameter space extended to include the mesoscale eddy closure and forcing masks in the OBS_9D optimization, where Veros MLD biases are reduced by up to 23%. The most significant improvement takes place in the Southern Ocean as a result of decreasing the zonal wind stress by about 10%. A key take-away message from our study is that OGCM parameters must be tuned together to account for their strongly coupled impact on the state of the model.

7 Open Research

All model simulations in the manuscript have been done using Veros version 1.5.3 (Häfner et al., 2018, 2021). The model is available to download on github: `https://github.com/team-ocean/veros` and v1.5.3 is preserved on Zenodo (Häfner et al., 2023). Veros documentation is available on Zenodo as well (Häfner et al., 2023). The optimizer VerOpt (Stoustrup, 2021) is available on github: `https://github.com/aster-stoustrup/veropt`. The mixed layer depth field used as a target in the optimization experiments OBS_TKE and OBS_9D is based on the climatology by (de Boyer Montégut, 2022). The observed depth of the 18°C isotherm used as a target in the OBS_9D experiment is based on World Ocean Atlas 2023 Database (Reagan et al., 2024). Veros model output presented in this paper is available at `https://sid.erda.dk/cgi-sid/ls.py?share_id=dGN0t2pSbf`.

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Appendix A Veros Parameterizations

The **TKE** scheme is a 1.5-order turbulence closure commonly used in ocean models, such as Nucleus for European Modelling of the Ocean (NEMO, Madec & the NEMO team, 2016) included in the Ocean Model Intercomparison Project (OMIP, Griffies et al., 2016). Eddy viscosity K_m is parameterized as:

$$K_m = c_k l_k \bar{e}^{1/2}, \tag{A1}$$

where l_k is the mixing length scale, c_k is the mixing coefficient and $\bar{e} = \frac{1}{2}(\overline{u'^2 + v'^2 + w'^2})$ is the turbulent kinetic energy. The zonal, meridional and vertical velocities are symbolized by u, v and w. The overbar and apostrophe indicate the mean and fluctuating components in Reynolds decomposition, respectively: $u = \bar{u}+u'$. The prognostic equation for $\bar{e}$ is:

$$\frac{\partial \bar{e}}{\partial t} = \frac{\partial}{\partial z}\Big(K_e \frac{\partial \bar{e}}{\partial z}\Big) - c_\epsilon \frac{\bar{e}^{3/2}}{l_\epsilon} - K_h N^2 + K_m Sh^2, \tag{A2}$$

where $K_e = \alpha_{tke} K_m$ is the vertical diffusive flux of $\bar{e}$, $K_h = K_m P_{rt}^{-1}$ is the eddy diffusivity with Prandtl number P_{rt}, N is the buoyancy frequency, $Sh^2 = (\frac{\partial \bar{u}}{\partial z})^2 + (\frac{\partial \bar{v}}{\partial z})^2$ is the vertical squared shear, and l_ϵ is the dissipation length scale. Aside from the three parameters α_{tke}, c_k and c_ϵ, the model allows for setting a limit for minimum eddy viscosity and diffusivity: min_{K_m} and min_{K_h}, respectively.

The computation of the length scales l_k and l_ϵ in Veros follows Blanke and Delecluse (1993):

$$l_k = l_\epsilon = 2^{1/2} \bar{e}^{1/2} N^{-1}. \tag{A3}$$

Furthermore, Prandtl number is dependent on the Richardson number Ri: $P_{rt} = 6.6Ri$, $P_{rt} \in [1, 10]$. Assuming homogeneous and stationary turbulence, it is possible to derive the critical Richardson number Ri_c as a function of the TKE parameters. In this regime, Equation (A2) becomes a balance between the buoyancy flux, shear production and dissipation:

$$K_m Sh^2 - K_h N^2 - c_\epsilon \frac{\bar{e}^{3/2}}{l_\epsilon} = 0. \tag{A4}$$

After rearranging the terms, Eq. (A4) becomes:

$$\frac{P_{rt}}{P_{rt} - Ri_c} Ri_c = \frac{6.6}{6.6 - 1} Ri_c = c_\epsilon^{-1} \bar{e}^{3/2} l_\epsilon N^2 K_m \tag{A5}$$

Recalling that $K_h = K_m P_{rt}^{-1}$ and using the length scale definition from Eq. (A3), we get $K_m = c_k 2^{1/2} \bar{e} N^{-1}$. We end up with Ri_c as a function of c_k and c_ϵ:

$$Ri_c \approx 1.7 c_k c_\epsilon^{-1}. \tag{A6}$$

Tuning the $c_k c_\epsilon^{-1}$ ratio is therefore equivalent to picking the critical Richardson number of the model.

In the **EKE** closure, fluctuations in velocity ($\mathbf{u} = (u, v, w) = \bar{\mathbf{u}} + \mathbf{u}'$) and buoyancy ($b = \bar{b} + b'$) are parameterized as:

$$\overline{\mathbf{u}_h' b'} = -K_{gm} \nabla_h \bar{b} \tag{A7a}$$

$$\overline{w' b'} = K_{gm} \frac{|\nabla_h \bar{b}|^2}{N^2} \tag{A7b}$$

$$K_{gm} = LU \tag{A7c}$$

The Reynolds stress terms are scaled by the scalar thickness diffusivity K_{gm} that varies spatially. Here L is the eddy length scale, which can either be the Rhines scale (L_{Rhi}) or the Rossby radius (L_{Ro}). U is the eddy velocity: $U = \sqrt{\overline{\text{eke}}}$. Buoyancy is defined as $b = -g\frac{\delta \rho}{\rho_0}$. The eddy kinetic energy budget is integrated as a prognostic model equation:

$$\begin{aligned}\frac{\partial \overline{\text{eke}}}{\partial t} &= K_{gm} |\nabla_h \bar{\mathbf{u}}|^2 + K_{gm} \frac{|\nabla_h \bar{b}|^2}{N^2} + \frac{f^2}{N^2} \frac{\partial K_{gm}}{\partial z} \frac{\partial \bar{E}}{\partial z} \\ &+ \nabla_h \cdot K_{gm} \nabla_h \overline{\text{eke}} + \frac{\partial}{\partial z} \left(0.1 K_{gm} \frac{f^2}{N^2} \frac{\partial \overline{\text{eke}}}{\partial z} \right) - \text{eke\_}c_\epsilon \frac{\overline{\text{eke}}^{\frac{3}{2}}}{L},\end{aligned} \tag{A8}$$

where f is the Coriolis frequency and $\bar{E}$ is the horizontal mean kinetic energy. In Eden and Greatbatch (2008), the following tuning choices are made:

- $L = min(2L_{Ro}, 0.3L_{Rhi})$
- $\text{eke\_}c_\epsilon = 0.1$

In Veros, an additional tuning parameter $\text{eke\_}c_k$ is introduced, such that $K_{gm} = \text{eke\_}c_k L \sqrt{\overline{\text{eke}}}$. Additionally, the turbulent vertical dissipation of eddy kinetic energy in the prognostic equation is scaled by eddy viscosity K_m instead of the $0.1 K_{gm} f^2 / N^2$ term.

Appendix B Additional error metrics

B1 Boundary punishment

Given the i^{th} parameter vector of length d, $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,d}]$, the boundary punishment term for the i^{th} objective function value in the OBS_9D experiment is calculated as:

$$\text{boundary punishment} = \frac{1}{d} \sum_{j=1}^{d} \epsilon_b^{i,j}. \tag{B1}$$

Here, $\epsilon_b^{i,j}$ is the boundary punishment of each individual parameter value at the point i. The boundary punishment varies between 0 and 1, $\epsilon_b \in [0,1]$, and its value decays exponentially away from the boundary. For a parameter $x_{i,j} \in [l_j, u_j]$,

$$\epsilon_b^{i,j} = \begin{cases} 1 & \text{if } x_{i,j} < l_j \text{ or } x_{i,j} > u_j \\ \exp(-50\frac{x_{i,j}-l_j}{u_j-l_j}) & \text{if } x_{i,j} < c_j \\ \exp(-50\frac{u_j-x_{i,j}}{u_j-l_j}) & \text{if } x_{i,j} > c_j, \end{cases} \tag{B2}$$

where $c_j = 0.5(l_j + u_j)$ is the center of the $x_{i,j}$ parameter range. The high value of the decay rate is chosen such that the boundary punishment term does not contribute significantly to the total objective function value unless $x_{i,j} \approx l_j$ or $x_{i,j} \approx u_j$.

This approach works well for all parameters that are linearly distributed within their parameter space. However, in the OBS_9D experiment, we reduce the bounds of min_{K_m} to span only across two orders of magnitude instead four, and subsequently drop the logarithm, ending up with $min_{K_m} \in [10^{-6}, 10^{-4}]$. For this set of bounds, the boundary punishment term exceeds 0.1 for $min_{K_m} < 5.6 \cdot 10^{-6}$ and $min_{K_m} > 0.954 \cdot 10^{-4}$. As a consequence, the range of min_{K_m} values with magnitude 10^{-6} will be disproportionately punished compared to min_{K_m} values with magnitude 10^{-4}, which is not the desired effect. The distribution of min_{K_m} is thus dominated by the boundary punishment term in the range $min_{K_m} \in [10^{-6}, 10^{-5}]$.

The results of the TWIN experiment indicate that MLD is relatively insensitive to min_{K_m} on the global scale, and the examination of various error sources as a function of min_{K_m} shows that this bug has a negligible effect on the optimization results.

B2 ACC and AMOC errors

The modeled ACC and AMOC transport strengths are computed using the horizontal stream function:

$$\Psi(x,y,t) = \int_D^0 \int_W^x v(x',y,z,t)dx'dz \tag{B3}$$

and the vertical stream function:

$$\hat{\Psi}(y,z,t) = \int_D^z \int_W^E v(x,y,z',t)dxdz', \tag{B4}$$

respectively. The ACC transport for each point i is estimated by the maximum $\Psi(x,y)$ for a box bounded by $x \in [64°W, 56°W]$ and $y \in [70°S, 62°S]$. The AMOC transport for each point i is estimated by the maximum $\hat{\Psi}(y,z)$ within the bounds $y \in [24°N, 34°N]$ and $z \in [500\ m, 2000\ m]$.

The target value for the acceptable range of the ACC strength is determined using two references: Cunningham et al. (2003) (136.7±7.8 Sv) and Donohue et al. (2016) (173.3 ± 10.7 Sv). Taking 136 Sv as the lower bound and 173 Sv as the upper bound, we get $\text{ACC}_{target} \pm \sigma_{\text{ACC}} = 155 \pm 19$ Sv. For the AMOC strength, we use 17.6±3.6 Sv from Fu et al. (2020) rounded up to $\text{AMOC}_{target} \pm \sigma_{\text{AMOC}} = 17.6 \pm 4$ Sv.

The errors ACC_{err} and AMOC_{err} vary between 0 and 1. If the modeled transport falls within the acceptable range (within one σ from the target value), the error is set to zero. If the modeled transport falls into unacceptable range (more than $n\sigma$ away from target, where $n = 2$ for the AMOC and $n = 3$ for the ACC), it is set to 1. In between the two extrema, the error grows exponentially:

$$F_{err}(\delta_i) = \begin{cases} 1 & \text{if } \delta_i > n\sigma \\ \frac{1}{\mathrm{e}-1}\left(\exp\left[\left(\frac{\delta_i-\sigma}{n\sigma}\right)^2\right] - 1\right) & \text{if } \sigma < \delta_i < n\sigma \\ 0 & \text{if } \delta_i < \sigma, \end{cases} \tag{B5}$$

where δ_i is the absolute difference between the target and the i^{th} simulated transport strength.

B3 18C error

The depth of the 18°C isotherm D is computed from observations and model output using:

$$D = \frac{D_{min} + D_{max}}{2}, \tag{B6}$$

where $T(D_{min}) = 18+2\sigma_T$ and $T(D_{max}) = 18-2\sigma_T$, with temperature T and uncertainty σ_T. We use the WOA23 temperature observations to compute the target D, D_{min} and D_{max}. For the models, we set $\sigma_T = 0.1$ °C. For both the observations and the model output, the depth of the 18°C isotherm is zonally averaged. The error:

$$18\mathrm{C}_{err} = \frac{1}{N}\sum_{j=0}^{N} \epsilon_j, \tag{B7}$$

where N is the number of grid cells within the latitudinal span $y \in [60°S, 60°N]$ and

$$\epsilon_j = \begin{cases} 0 & \text{if } |D_{target} - D_i| < \delta \\ \frac{\min[(D_i - D_{min}),(D_{max} - D_i)]}{D_{target}} & \text{if } |D_{target} - D_i| > \delta, \end{cases} \tag{B8}$$

with $\delta = \frac{1}{2}(D_{max} - D_{min})$. I.e., if the simulated depth of the 18°C isotherm falls within the range $[D_{min}, D_{max}]$, $18\mathrm{C}_{err}$ is zero, otherwise it is equal to the absolute error relative to the target 18°C isotherm depth.

As we analyzed the results of the OBS_9D experiment, we noticed that the $18\mathrm{C}_{err}$ is equal to zero for some of the simulations with the highest objective function values. We found a bug in the code used to compute the $18\mathrm{C}_{err}$ where ϵ_j is set to zero for all D values less than D_{min}. In other words, the simulations for which the 18°C isotherm is too shallow were interpreted as error-free in our optimization sequence. We now discuss the consequences of this bug.

Figure B1 shows the objective function values $f_{\text{OBS\_9D}}(\mathbf{x})$, the objective function recalculated with the corrected $18\mathrm{C}_{err}$, and the difference between the two. Fortunately, in the vast majority of cases, the corrected $18\mathrm{C}_{err}$ is correlated with $f_{\text{OBS\_9D}}(\mathbf{x})$. This indicates that although the contribution from the corrected $18\mathrm{C}_{err}$ significantly impacts the magnitude of $f_{\text{OBS\_9D}}(\mathbf{x})$, it does not alter its distribution. Unsurprisingly, the $18\mathrm{C}_{err}$ is tightly correlated with the eke_c_k parameter value. When eke_$c_k > 1$, the isopycnals in the tropics are flattened, leading to a shallow bias in $18\mathrm{C}_{err}$. Notably, $18\mathrm{C}_{err}$ is minimized when eke_$c_k \approx 0.1$, further underlining the significance of eke_c_k for the modeled ocean state.

Acronyms

OGCM Ocean General Circulation Model
TKE Turbulent Kinetic Energy
MLD Mixed layer depth
GP Gaussian process
BO Bayesian optimization
EKE Eddy kinetic energy
MLL Marginal log likelihood
rmse Root mean square relative error

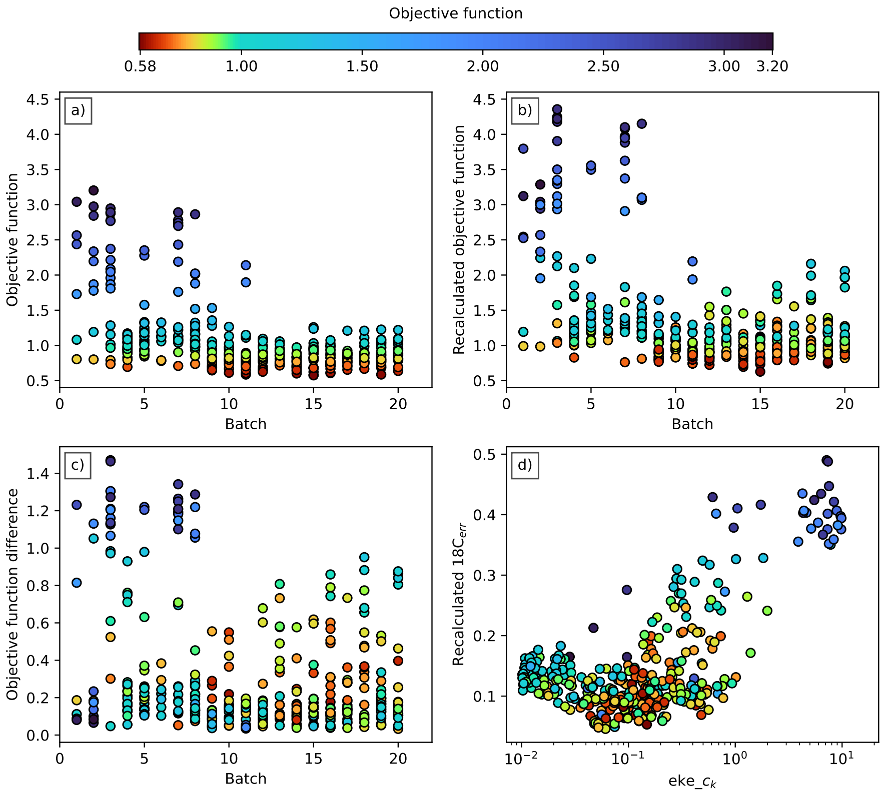

Figure B1. Overview of the effect of the 18C$_{err}$ computation bug on the objective function: a) the objective function calculated with 18C$_{err}$ in the OBS_9D optimization; b) the objective function recalculated with the corrected 18C$_{err}$; c) the difference between the original and corrected objective function values; and d) the recalculated 18C$_{err}$ as a function of eke_c_k.

Notation

Bayesian optimization

$\boldsymbol{f}(\mathbf{x})$ The objective function
$\boldsymbol{\mathcal{X}}$ The parameter space of the objective function
$\mathbf{x}$ An input point, $\mathbf{x} \in \mathcal{X}$
$\mathbf{x}_*$ A test point, $\mathbf{x}_* \in \mathcal{X}$
$\boldsymbol{X}$ A matrix of input points
$\boldsymbol{X}_*$ A matrix of test points
$\mathbf{f}(\boldsymbol{X})$ A vector containing objective function evaluations
$\mathbf{x}^*$ The global minimum of the objective function
$\boldsymbol{\chi}^*$ The minimum of evaluated (input) points: $\chi^* = \arg\min \mathbf{f}(X)$
$\boldsymbol{K}(\boldsymbol{X}, \boldsymbol{X}')$ The Gram matrix; each element $k_{i,j} = k(\mathbf{x}_i, \mathbf{x}'_j)$
$\boldsymbol{\mathcal{N}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ Normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix Σ
$\mathbf{q} \sim \boldsymbol{\mathcal{N}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ The vector $\mathbf{q}$ is drawn from the normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix Σ
$\boldsymbol{\mathcal{GP}}(\boldsymbol{m}(\cdot), \boldsymbol{k}(\cdot,\cdot))$ Gaussian process defined by mean $m(\cdot)$ and kernel $k(\cdot,\cdot)$
$\boldsymbol{g} \sim \boldsymbol{\mathcal{GP}}(\boldsymbol{m}(\cdot), \boldsymbol{k}(\cdot,\cdot))$ The function g is a sample Gaussian process with mean $m(\cdot)$ and kernel $k(\cdot,\cdot)$
$\boldsymbol{\theta}$ GP model hyper-parameter vector
$\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\mu}}(\mathbf{x}), \hat{\boldsymbol{\sigma}}(\mathbf{x})$ The hyper-parameter vector, mean and standard deviation of the GP model for which the MLL is maximized

Oceanography

$\bar{\cdot}$ Mean component in Reynolds decomposition
$\cdot'$ Turbulent component in Reynolds decomposition
$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$ Geographical coordinates: zonal, meridional and vertical direction
$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ Three-dimensional flow: zonal, meridional and vertical velocity
$\boldsymbol{\rho}$ Sea water potential density
$\boldsymbol{\delta\rho}$ Potential density difference
$\boldsymbol{T}$ Sea water temperature
$\boldsymbol{z}_{\mathbf{ref}}$ Reference depth for calculating MLD
$\boldsymbol{K_m}$ Eddy viscosity
$\boldsymbol{min_{K_m}}$ Minimum eddy viscosity
$\boldsymbol{K_h}$ Eddy diffusivity
$\boldsymbol{min_{K_h}}$ Minimum eddy diffusivity
$\boldsymbol{e}$ Turbulent kinetic energy
$\boldsymbol{l_k}$ Mixing length scale
$\boldsymbol{l_\epsilon}$ Dissipation length scale
$\boldsymbol{N}$ Buoyancy frequency
$\boldsymbol{b}$ Buoyancy
$\boldsymbol{Sh}$ Vertical shear
$\boldsymbol{c_k}$ Mixing coefficient in $K_m = c_k l_k \bar{e}^{1/2}$
$\boldsymbol{c_\epsilon}$ Dissipation coefficient in $\epsilon = c_\epsilon l_\epsilon^{-1} \bar{e}^{3/2}$
$\boldsymbol{\alpha_{tke}}$ Vertical diffusive flux coefficient in $K_e = \alpha_{tke} K_m$
$\boldsymbol{Ri}$ Richardson number $Ri = N^2 Sh^{-2}$
$\boldsymbol{Ri_c}$ Critical Richardson number
$\boldsymbol{\gamma_{R_f}}$ Mixing efficiency coefficient
$\boldsymbol{P_{rt}}$ Turbulent Prandtl number
$\boldsymbol{K_{gm}}$ Mesoscale eddy diffusion coefficient
$\boldsymbol{K_{gm}}$ Isopycnal tracer diffusion coefficient
$\mathbf{eke\_}\boldsymbol{c_k}$ EKE mixing coefficient

eke_c_ϵ EKE dissipation coefficient
c_{τ_x} Zonal wind stress coefficient
c_{τ_y} Meridional wind stress coefficient
α_s Surface salinity mask coefficient

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